

The Value of Virtual Engagement: Evidence from a Running Platform

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Abstract

Fueled by remarkable developments in mobile and wearable technologies coupled with the power of the Internet, a multi-billion industry constructed on digital platforms, *Connected Fitness*, has emerged. Sustained engagement is a crucial prerequisite for healthy behavior, and platforms that offer connected fitness services must contend with the challenge of engaging new users. In this study, we examine whether a virtual race, an event in which runners can participate remotely in an organized activity, can serve as an effective strategy to amplify users' long-term engagement related to healthy behavior. We study the impact of offering virtual races to runners on a digital fitness platform on their engagement with the platform and level of physical activity. Leveraging the match of the theme of the virtual races and users' zodiac signs as the instrument to support causal inference, we find that a virtual race increases the engagement of new users by 42.8 days and total spending on related e-commerce on the platform by 48.4% during our study period. We also find that the positive impact of virtual races transcends the typical mechanism of social connections: results show virtual races can increase the engagement with the platform even when their social connections are low. Finally, we find that existing users who participate in a virtual race outrun those in the control group by 4.41 km in a three-week period. Our study highlights the potential of new forms of digital interactions, powered by mobile and wearable technologies, on user behavior in fitness platforms.

Keyword: virtual race, digital platform, user engagement, social interaction, healthy behavior

1. Introduction

In recent years, fueled by remarkable developments in mobile and wearable technologies coupled with the power of the Internet, a multi-billion industry, *Connected Fitness*, has emerged¹. Players in this sector rely not only on the features of mobile applications, but also seek to leverage the capabilities of digital platforms, including community and social functions, to engage users in physical activities in a more robust manner. Numerous vendors have joined the fitness virtualization bandwagon. Peloton, an American exercise equipment company, attracted 5.4+ million users in its virtual community² as of March 31, 2021;

¹ <https://www.globenewswire.com/en/news-release/2020/11/24/2132517/0/en/Global-fitness-app-market-size-valuation-to-reach-USD-15-96-billion-by-2026.html>

² <https://investor.onepeloton.com/>

Nike accumulated more than 170 million users to participate in online Train and Nike Run Club³; and Apple launched Fitness+ workout service to enrich its Apple Watch and iPhone ecosystem⁴. Market research suggests strong growth⁵: the global online (virtual) fitness market is projected to increase sharply from \$6.04 billion in 2016 to \$59.2 billion by 2027.

At the core of the connected fitness movement is the underlying digital platform that virtualizes social interactions. Indeed, recent research conducted by J.P. Morgan⁶ emphasizes the importance of building platforms that integrate social factors and modern hardware (treadmills, wearable devices, etc.) to form communities for a healthy life. For fitness platforms such as those for virtual racing to succeed, a key challenge is to retain and engage users. This challenge has been identified as a strategic imperative for platform businesses in general (DellaVigna and Malmendier 2006; Hu 2005; Van den Poel and Lariviere 2004), and is further amplified for connected fitness platforms, because their central offerings are activities (e.g., biking, running, yoga) that are effortful in the long run for users (Milne-Ives et al. 2020). Forming new healthy habits is notoriously challenging for newcomers because it involves behavioral modifications. Charness and Gneezy (2009) emphasized the importance of overcoming the “initial resistance” to behavioral changes, given the huge startup costs. Thus, passing the threshold of habit formation is necessary for both sustained healthy behaviors, and for continued engagement with the platform.

Because virtual races allow individuals to engage in a collective experience that can scale to unprecedented levels, surpassing the reach of both online support communities and direct offline peer support, they can foster camaraderie and connection across large numbers of participants. In this paper, we examine whether virtual racing, an important emerging form of connected fitness, can serve as an effective strategy to overcome startup resistance and positively influence long-term engagement related to physical

³ <https://www.modernretail.co/retailers/to-grow-its-direct-business-nike-is-looking-to-get-more-out-of-its-app-users/>

⁴ <https://www.cnn.com/2020/12/18/cnn-underscored/apple-fitness-plus-review/index.html>

⁵ <https://reports.valuates.com/market-reports/QYRE-Auto-37C4868/global-virtual-fitness-hardware>

⁶ <https://www.jpmorgan.com/insights/research/digital-fitness-industry>

activities on digital fitness platforms⁷. The answer to this question is not immediately obvious. The need for social interactions in any activity imposes requirements on the context where the activity will occur, with offline and online settings characterized by strengths and limitations. The advantages of being online such as scalability and accessibility, notwithstanding, there are worries that the virtual environment is deficient in its ability to offer the high-touch interactions that create value for users in offline settings. Critics argue that a virtual event simply cannot reproduce the atmosphere of offline activities due to the lack of face-to-face connections. For instance, some have expressed concerns over virtual races: “part of the reason why runners pay a fee for races is the camaraderie and environment⁸.” Although virtual races have lower participation costs and could, therefore, attract more users, it is also plausible that users’ sense of belonging may be adversely affected when events are moved online. This may dampen the user experience, negatively influencing user participation in physical activities and the loyalty of platform members, especially for new users.

Despite the debate, there is a limited number of empirical studies of the impact of virtualization on user engagement and physical activities, a void we aim to fill. We partnered with a leading physical activity platform in Taiwan that provides free virtual races for runners. We collected data on 18 virtual races involving 44,391 newly registered runners to estimate their effects on participants’ engagement and 37,048 existing users to examine effects on physical activity. Given that new users play a critical role for business growth (Dagger and Danaher 2014) and it is more challenging for new users to overcome the initial resistance to behavioral change, our work focuses mainly on the effects of a virtual race on *new* users’ long-term engagement. We further provide in-depth insights quantifying a virtual race’s effect on performance (physical activities) outcomes for existing users.

Our empirical analyses yield several novel insights. We find that new platform users who participate in a virtual race that is closest to their registration date increase their engagement on the platform

⁷ We use the phrase ‘user engagement’ to reflect retention, which is one of the most important measurements for mobile apps and online platforms. We examine the robustness of this operationalization in section 4.5 with additional analyses using other dependent variables, such as log-in frequency and spending, to represent user engagement.

⁸ <https://www.washingtonian.com/2020/05/22/how-is-the-pandemic-affecting-the-running-race-industry/>

by 42.8 days, and total spending on related e-commerce offered on the platform by 48.4%, providing strong evidence that a virtual race is as an effective means to engage newcomers on physical activity platforms. We then investigate the effects of participation in a virtual race on an existing user's running distance in a three-week period after the virtual race. We find that individuals who participate in a virtual race outrun those in the control group in the post-treatment three-week period by 4.41 km, representing an 8.7% increase in performance.

We conduct several additional analyses to uncover the mechanisms underlying the enhanced engagement. First, by exploring the precise nature of social interaction in the virtual race setting in greater depth, we show that virtual races can be especially beneficial for individuals who incur higher costs of seeking social interactions or individuals who have a stronger desire to seek social interactions. Analysis of heterogeneous effects shows that these two types of users engage longer with the platform, indicating that virtual races may be effective in fulfilling social needs. Second, we show that virtual races can substantially help individuals build social connections, especially during the 9-day race challenge period. Virtual race participants experience a 64% increase in social connections as compared to non-participants. Considering the evidence regarding many other types of online events that failed to meet initial expectations, this finding is particularly important for virtualizing events that involve strong social interactions and helps mitigate a potential concern that community social interactions afforded by offline racing cannot be replicated. We further find that the experience of participating in a virtual race increases individuals' likelihood to participate in future virtual races and offline races, suggestive of an increase of interest in marathon races. Taken together, these findings provide strong evidence of the extent to which virtualizing a traditional physical event is effective in promoting healthy behavior.

Our work contributes to the growing stream of literature that examines how mobile and wearable devices can be used to increase individuals' physical activities and other health outcomes (Adjerid et al. 2021; Ghose et al. 2021; Kato-Lin et al. 2016; Liu et al. 2021). We focus on virtualization of physical

exercise, which, despite its prevalence and continued growth, has received little attention in research⁹. Our findings provide important managerial insights on how the connected fitness industry can leverage virtualization to enhance user loyalty and form healthy habits.

2. Conceptual Foundations and Research Questions

We provide the background for the study, focusing on the emerging virtualization of physical activities. Because virtual racing is fundamentally an application of digital technologies for health and wellness, we present a review of relevant literature on digital health in the information systems and healthcare domains. Following this, building on these conceptual and empirical foundations, we propose research questions regarding the impacts of virtualization of marathon events on user behavior.

2.1. Background

Offering a new way to compete with other runners and with significant inherent flexibility, virtual racing is increasingly becoming popular. In a virtual race, runners can participate remotely in racing events hosted by the digital platform. The virtual race relies heavily on modern digital technologies: runners typically run a set distance using a dedicated smartphone app or GPS tracker to record the running in any location at dates specified by event organizers. A recent report indicates that there are 32.5 million people in China who participated in virtual races in 2018 alone¹⁰, and even Walt Disney sponsors an annual 5k virtual challenge that attracts tens of thousands of participants¹¹. The COVID-19 pandemic elevated the importance of virtual racing, with many marquee running events transitioned to a virtual format. For instance, for the first time in its 124-year history, the Boston Marathon, the world's oldest annual marathon, was canceled in 2020 in response to the concerns about the spread of COVID-19 and held as a virtual race. With the advancements in VR and AR, the trend of virtualization of sports events is expected to remain

⁹ There are some exceptions. For instance, Berkowitz et al. (2021) explore how an “educational peloton” can create a safe and protective community to drive engagement and work satisfaction.

¹⁰ <https://www.athensauthenticmarathon.gr/site/index.php/en/news/news-articles/83-news-2019/599-aims-gala-2019>

¹¹ <https://www.rundisney.com/events/virtual/virtual-running-series/>

strong post-pandemic¹². We list a few platforms or mobile applications in Table B1 (Online Appendix) to briefly showcase the popularity of virtual races and the frequency with which these events are hosted.

Our setting is that of a running platform that hosts marathons, a race whose official distance is roughly 26.2 miles (42 km). Thousands of marathons are held all over the world every year, attracting both competitive runners and recreational athletes. Large marathon races can have tens of thousands of participants (Association of International Marathons and Distance Races 2016). Cities that host marathons typically leverage marathon races to attract people from all over the world as a means of revenue generation for local businesses (Papanikos 2015; Saayman and Saayman 2012). Most runners, especially recreational athletes, do not aim to win the competition; rather, these runners generally seek to either break time barriers, improve their own best records, or simply for the satisfaction of completing a difficult physical challenge.

The rigorous training regimen typically required to complete a marathon allows individuals to build endurance and form exercise habits (Yeung et al. 2001). The benefits of habitual exercise are well known: regular exercise can lead to a reduction in rates of coronary heart disease, depression, and obesity (Thompson et al. 2003). The popularity of running as a form of exercise has risen in the past two decades; it is among the most practiced sports worldwide¹³. Recent data¹⁴ suggest that there were 18.1 million people registered for U.S. road races in 2018. In Asia, the setting for our study, running has become extremely fashionable in recent years, with the number of marathon races increasing rapidly (Meyers 2017). Interestingly, analysis of 107.9 million race results from more than 70,000 races held between 1986 and 2018, suggests that the motivations to participate in a race have changed from being achievement-focused to those that are more psychological, health, and socially focused (Andersen 2019).

In contrast to a physical marathon, in a virtual one, participants can run the race at different locations and at different times. Typically, participants need to use a specific mobile application provided

¹² <https://www.globenewswire.com/news-release/2021/10/20/2317515/0/en/COVID-19-Emergence-Insight-Global-Augmented-Reality-in-Healthcare-Market-to-Hit-1-918-6-Million-by-2026-Growing-at-a-CAGR-of-25-5-from-2019-to-2026-Exclusive-Current-and-Forecast-O.html>

¹³ <https://www.statista.com/topics/1743/running-and-jogging/>

¹⁴ https://runningusa.org/RUSA/News/2019/Running_USA_Releases_2019_U.S._Running_Trends_Report.aspx

by the race organizer (generally, an online platform) to record their mileage. They can earn a digital completion proof after completing the required distance. Online platforms that support virtual marathons generally provide a variety of features that simulate the experience of a physical race, including numerical, vocal, and visual functions. The statistics of all runners, such as running time and detailed pace, are made available for ranking and cross-user comparison. Features include those that provide live interactions when users start running, such as cheering or reporting of completed miles, to enhance the running experience. Visual features displayed on a screen or VR headset are also used to make virtual races more vivid.

2.2. The Challenges of User Retention and Technology Mediation in Fitness and Health

User retention (or continued engagement) is a core challenge across diverse domains, with prior research identifying a variety of influences on retention, including customer satisfaction (Bolton 1998), switching costs (Chen and Hitt 2002, Ray et al. 2012), self-service enabled in online channels (Hitt and Frei 2002, Campbell and Frei 2010, Xue et al. 2007), personal service and self-service (Scherer et al. 2015), variety seeking behavior (Berné et al. 2001, McAlister 1982, McAlister and Pessemier 1982), and even the customer acquisition process (Thomas 2001). Overcoming this challenge is especially important for new customers because they are critical in boosting business growth (Dagger and Danaher 2014), and practitioners expend tremendous resources in attracting new users, sometimes even at the cost of existing users (Rosenberg and Czepiel 1984).

The retention challenge is particularly serious for digital platforms, especially those in the healthcare domain. For instance, Baumel et al. (2019) found the median daily active user rate among 93 qualified apps for mental health was only 4.0%, with breathing exercise apps the lowest at only 1.6%. Motivating users for behavior modification, such as physical exercise, is extremely difficult, especially after the “treatment period.” In one study on long-term sustained gym attendance, Carrera et al. (2020) focused on nonmembers of the gym and found a significant drop in gym visits after incentives were removed. DellaVigna and Malmendier (2006) documented the inconsistency between people’s choices and actual behaviors – many people choose to pay a membership fee in advance of exercise and end up paying more than paying a per-visit fee due to the actual fewer visits than expected, driven by the human propensity to

mispredict behavior. Acland and Levy (2015) recruited non-gym attendees and tested their pre-treatment prediction on future gym attendance, finding discordance by a factor of 2.5 to 5.5, suggesting a significant overestimation by new users about their ability to sustain exercise. Misconceptions about physical exercise are not only limited to overconfidence in continuing attendance, but also the overestimation of initial setup cost. In a deep exploration of users' long-term healthy behaviors, Charness and Gneezy (2009) acknowledge that overcoming the initial resistance is one of the most important factors in forming healthy habits (Laibson 1997; O'Donoghue and Rabin 1999). Interventions helping users pass the threshold of the initial setup cost seem to be effective (Beshears et al. 2021); however, pure extrinsic interventions like financial incentives might destroy people's intrinsic motivation via the crowding-out effect (Gneezy and Rustichini 2000).

The virtualization of exercise, including racing, that relies on the integration of modern technologies (e.g., mobile apps, wearable devices) and socially connected communities, might help runners to overcome the initial setup cost without crowding out their intrinsic motivations for forming healthy physical activity habits. Additionally, the socialization enabled by virtualization might motivate long-term engagement. We briefly review prior work at the intersection of mobile health and social networks (see Table B2 for a literature review). Multiple studies have documented the potential health benefits of mobile and social technologies across general physical activities as well as for specific disease conditions. Kato-Lin et al. (2016) find that the mobile-based visual diary is effective in improving user engagement with healthy eating, while Müller et al. (2016) find SMS text messaging can promote exercise in older adults. In the context of a mobile app for diabetes management, Ghose et al. (2021) find positive health outcomes in both short-term (e.g., blood glucose and glycated hemoglobin levels) and longer-term metrics (e.g., hospital visits and medical expenses) after the adoption of a mHealth app. Adjerd et al. (2021) show that a combination of wearable tools and gain-loss incentives can increase individuals' physical activities. Kwon et al. (2022) uncover the effect of mobile app usage in enabling self-regulation, which contributes more than offline office visits to the long-term sustainability of users' health.

For social elements in virtual settings, prior literature has documented several mechanisms as effective in influencing health outcomes. Yan and Tan (2014) find that patients can benefit from learning from others in an online community, showing that although information support is one of the most prevalent types in the online health community, emotional support is also effective in helping patients move to a healthier state. Huang et al. (2019) find that all three dimensions of social capital (i.e., structural, relational, and cognitive) positively increase the provision of emotional support, but structural and relational social capital facilitate engagement in companionship; and only cognitive social capital increases the provision of informational support. Aral and Nicolaides (2017) show that daily exercise is socially contagious, and Liu et al. (2021) find that the effect of reciprocity-based incentives for physical activity on a digital platform is contingent on the social closeness between users.

Despite the potential benefits they offer, prior research has also reported the existence of negative or non-significant effects in contexts utilizing digital technologies or social networks for health and physical activities. In a field experiment conducted in Uganda aimed at reducing risky health behavior, Jamison et al. (2013) found that a text message-based service providing sexual education led to even more risky sexual behaviors. Uetake and Yang (2020) documented a potential negative impact of peer influence on weight loss. Indeed, a systematic review of mHealth on chronic disease management by Hamine et al. (2015) showed that only 56% of studies significantly improved adherence behaviors and called for more research in this domain.

Our work contributes to this literature by focusing on the virtualization of fitness events that predominantly occur in an offline context: marathon races. The growing popularity of this activity fueled by advances in digital technologies combined with social networks and online communities underscores the need for a deeper understanding of outcomes for both users who participate and the platforms that offer these services. Our work is among the first to directly quantify the effect of virtual socialization of fitness events on individuals' physical activities and engagement.

2.3. Research Questions

The virtualization of activities characterized by high-touch offline interactions has been studied in areas such as work-from-home (WFH) and online learning. In these two domains, findings regarding the effects of virtualization are equivocal. For instance, while previous results show that WFH leads to an increase in performance (Bloom et al. 2015; Choudhury et al. 2021), in 2017, IBM, a pioneer of remote work, canceled its decade-long WFH policy and called thousands of employees back to the office, citing a need for better productivity, efficiency, and even innovation (Kessler 2017). For online learning, studies indicate that the course completion rate is lower in the online format than it is for a traditional face-to-face course (Hart et al. 2018; Murphy and Stewart 2017). Some studies show that many students are ill-prepared for online courses, leading to worse grade performance (Urtel 2008). Protopsaltis and Baum (2019) conclude that online education does not meet its initial expectations for reducing performance gaps among students.

Researchers have invoked many explanations for why some types of experiential offline events may not be amenable to virtualization. These include technical concerns (Aragon and Johnson 2008), lack of physical interaction (Pigliapoco and Bogliolo 2008; Tello 2008), and poor time management and locus of control (Holder 2007; Morris et al. 2005). Since racing is an activity that is highly experiential, a lack of physical interaction might create additional challenges that could undermine the effectiveness, causing individuals to exit from the platform, and resulting in a backfire effect that is harmful to retention.

As a counterpoint to this argument, it is plausible that with the help of modern technologies and social networking features, a virtual race might be able to effectively alleviate such concerns. This is likely to be true for at least five reasons. First, GPS-enabled smartphones or wearable devices are now ubiquitous and sophisticated enough to host a dedicated app from the online platform that provides easy-to-use tracking and other functions. Studies have shown that wearable devices can be an effective and popular means to increase physical activities (Adjerid et al. 2018; Hydari et al. 2022). Over the years, users are also becoming more comfortable and competent in using mobile apps and devices. Therefore, technological concerns are likely not major issues that cause negative feelings among runners, potentially resulting in an exit from the online platform.

Second, traditional races often involve a significant number of people running together, accompanied by an enthusiastic audience providing encouragement. By comparison, participants in a virtual race mostly run alone or with only a few friends. However, despite the loss of in-person interactions, a virtual race, by leveraging social networking features, can provide an alternative means to enable social interactions. GPS and smartphone technologies can feed data to the platform to create a common experience for participants. With runners' profiles and running records easily accessible and publicly available in an online platform, interacting with previously unknown runners is much easier in a virtual race than in an offline one. Thus, although the social interactions in virtual races are different from traditional races, attending a virtual race could allow participants to enjoy more online social interactions, thereby amplifying their loyalty to the platform. Indeed, prior research in the online platform context has suggested that socializing new users is the key to engagement and involvement. For instance, Ciampaglia and Taraborelli (2015) show that a lightweight socialization tool in Wikipedia can increase newly registered users' retention. Likewise, Morgan and Halfaker (2018) argue that effective socialization is vital for users' sustainability, and show that Wikipedia Teahouse, a space that allows new editors to interact with experienced editors, can increase newcomers' engagement. Chen et al. (2017) demonstrate that reciprocity and peer recognition play a significant role in user engagement in online communities. Joyce and Kraut (2006) find that newcomers who get a reply on online communities are 12% more likely to post again than those who do not get a reply. Thus, virtual races might socialize newcomers in the platform and, in turn, increase their engagement, which subsequently promotes the performance of the physical activity via stronger social support (Eyler et al. 1999; Springer et al. 2006).

A third reason why a virtual race may be efficacious is that common experience enables individuals to build emotional connections within the community. Studies have shown that common experience is important for engagement in social movements (Bunnage 2014), and can foster a sense of community (López 2013). Although most participants run by themselves, the virtual environment also provides features to negate the feeling of running alone. For example, individuals can share their running experience with others, and receive encouragement and helpful feedback. Such sharing and interactions might help to

increase the bond with the platform. Research suggests that providing individuals with historical reminders can prompt them to consider their past behaviors, making them engage more with the platform (Cheshire and Antin 2008). Therefore, the common experience with other runners may amplify users' loyalty to the platform and engagement, with shared experience occasionally reminding participants of good memories.

Fourth, self-management (Breevaart et al. 2014), especially time management (Nawrot and Doucet 2014) is a key to improve engagement. Virtual races remove certain costs and constraints, affording users greater flexibility in optimizing their time. Compared to traditional offline races that are typically held in a circumscribed time window, virtual races provide a longer time-window to complete the race (e.g., 2 days (during the weekend), 7 days (1 week), or even 1 month). In addition, the flexibility afforded by the elimination of cutoff times (typically, 6 hours for a full marathon) in virtual races allows runners who may otherwise not have the fortitude to complete an offline race to be able to do so in the virtual setting. Moreover, motivated by peer influence from other runners in the virtual race who have completed early, runners might be able to learn from others' experience and build the strength and commitment to complete races that they typically may not otherwise finish themselves.

Finally, online virtual racing is similar in some ways to other features offered on digital platforms, such as points, badges, rewards, leaderboards, or feedback whose usage has been escalating in recent years (Lister et al. 2014; Looyestyn et al. 2017) and that have been applied to many health-related contexts, including physical activity, diet and weight loss, personal hygiene, and medication (Pereira et al. 2014). However, a virtual race is also different from points, or rewards that provide mostly self-management or from leaderboards, feedback from online friends, that leverage social comparison or social support to motivate behavior change (Bojd et al. 2022; Edwards et al. 2016; Looyestyn et al. 2017; Yan and Tan 2014). For many of these (social) features to be effective, it is essential to build social connections, which is notoriously challenging for newcomers. This creates a paradox: platforms want to use some of these features to engage users and increase their online friends, but many of these features suffer from the cold start problem. In contrast to what is needed for optimal use of these features, a virtual race represents an event with a common experience. Therefore, even without building substantial social connections ex-ante,

virtual races may allow participants to experience a sense of social support when other (unknown) users leave comments or press “like” for the completed virtual race, or make either upward or downward comparisons when participants check other publicly available running records.

In short, powered by modern wireless technologies and digital platforms, virtual races have the potential to jump start a new member’s social support and shared experience, elements that have been shown to be vital for engagement (Bunnage 2014; Ciampaglia and Taraborelli 2015). Following these arguments, we pose the following research question: *Does participation in a virtual race increase new users’ engagement with the platform?*

By design, a virtual race requires participants to run a pre-determined distance (e.g., 10 km, 21 km, 42 km) in a fixed time window. Therefore, we should expect to see a boost in physical activity within the window. However, physical activity takes time and effort to complete; and if virtual races yield negative experiences for participants, it might discourage exercise in a subsequent time period. Extending from our first research question, we also aim to address the following: *Does participation in a virtual race positively affect individuals’ physical activity after the virtual race?*

3. The Impact of a Virtual Race on New Users’ Platform Engagement

3.1 Research Context

We collected data from a platform that offers tracking and social networking for runners, henceforth referred to as *RunningPlatform* to maintain anonymity. *RunningPlatform* is a leading physical activity website in Taiwan that, at the time of data collection, had more than 300,000 registered users. *RunningPlatform* provides a free mobile application for individuals to track their physical activity. It also supports various types of wearable devices that can be synchronized with the platform. Most users rely on the platform’s free mobile application or wearable devices to track their physical activities. Advanced GPS technology allows users to view their route in detail for each running episode. The platform captures detailed information for each running episode, including duration, average heart rate, temperature, and humidity. Moreover, the platform offers standard social networking functions. For instance, individuals can

follow or be followed by other runners, and see others' physical activity if their running records are publicly available.

Figure A1 shows several key features the platform provides, of which enabling users to track their physical activities is central. One can use the mobile application or wearable devices to track running episodes and monitor exercise through various summary statistics. In addition, as indicated, one can follow or be followed by other runners, and users are able to interact with each other by pressing “likes” or leaving “comments” on friends' running posts. Moreover, the platform provides an e-commerce feature that allows runners to purchase products through the mobile application. The app also provides several useful features, such as ranking runners via their monthly performance, allowing runners to set monthly goals, providing a list of official marathon races, offering weight management control, and allowing users to form running clubs, etc.

In addition to the free virtual races provided on its online platform, the company allows individuals to purchase a race medal at an affordable price (roughly \$3 each). Each virtual race has exactly the same rule and is “themed” with a zodiac sign (e.g., Pisces Virtual Race, Gemini Virtual Race). Figure 1 shows a sample screenshot of registration for the Cancer Virtual Race in 2018. For each race, individuals have a 9-day window to complete the race, which always starts on a Saturday and ends on the next Sunday. This time window provides sufficient flexibility to runners. There is no location constraint for the virtual race, so participants have the flexibility to run the race anywhere. They can choose to participate in one of the following racing challenges: 5 kilometers, 10 kilometers, half marathon (roughly 21 kilometers), and full marathon (roughly 42 kilometers). Before a virtual race starts, users are given a period of time to register for it. The zodiac theme implies that the gap between each race is roughly four weeks, with one zodiac-theme-based virtual race per month. The race medal is different for each game, and can be purchased during registration. Users are also able to buy other merchandise during registration, but there is no requirement to do so.

RunningPlatform's business model relies on the monetization of users through the sale of physical activity-related products on its own e-commerce site (e.g., shoes, T-Shirts, or socks). Thus, attracting and retaining customers is critical for increasing sales and revenue.

3.2 Data

Our data set includes eighteen virtual races organized by the platform over approximately 18 months, from December 16th, 2017, to May 19th, 2019 (representing the start and end dates of races)¹⁵. In total, we have 44,391 new users who registered on the platform during this period as our final sample. To differentiate our additional analysis in a later section, we label this data set of new users as *Sample 1*. We observe whether participating in a race that is closest to users' registration time significantly increases participants' engagement. We measure engagement by using individuals' log-in records that are maintained by the platform. We use the difference between the last log-in date and the registration date to calculate the number of days an individual engages on this platform (henceforth, "alive days"), which is a typical metric for a user's length of stay in a community or retention (Rossmann et al. 2016; Viljanen et al. 2016). The definition is also close to the concept of customer retention in marketing, which represents a customer's time of defection from an existing service (Nitzan and Libai 2011). In addition, "Alive days" (or retention rate) is one of the most important measurements for mobile apps and online platforms. For example, DAU (Daily Active Users) and MAU (Monthly Active Users), which measure the number of users are "alive" in a day/month, are two of the most reported and important key performance indicators that are monitored by different stakeholders, and even competitors (Frankel and Li 2004; Weitz et al. 2014). In our setting, we observe this when a user ceases to log into the platform. To check for robustness, we use the number of log-ins and users' spending as alternative proxies for user engagement.

Table 1 provides a detailed definition of all variables, while Table 2-1 shows summary statistics for each variable in *Sample 1*. On average, individuals stay engaged with the platform for 213.6 days. Half of our sample is female, the average user age is roughly 35 years old, and approximately 8% participate in

¹⁵ The data was collected prior to COVID-19; thus, users' registrations and participation in virtual races were not affected by external factors.

a virtual race shortly after registration¹⁶. We present the correlation information in Table 3-1. Participation in a virtual race is positively associated with the number of alive days on the platform. Male and older users have a higher inclination to stay on this platform. The correlation results must be interpreted with caution as we do not control for covariates.

3.3 Basic Empirical Models

We start with a parsimonious ordinary least square (OLS) model to examine the effect of participation in a virtual race on user engagement:

$$AliveDays_i = \alpha + \beta_1 \times VirtualRace_i + \Sigma \gamma_w \times Z_i + \eta_t + \varepsilon_i \quad (1)$$

where *AliveDays* denotes the number of days an individual *i* stays on this platform, measured by the difference between the last log-in date and the registration date; *VirtualRace* represents whether an individual *i* participates in a virtual race that is closest to the registration date or not; and *Z* is a vector of individual demographics including age, gender, weight, and height¹⁷. To account for unobserved factors that affect individuals' behaviors and vary over time, we add η_t as a fixed effect for each of the 18 virtual races in the model.

The dependent variable, *AliveDays*, is right-censored because we only observe the time of survival until the date of data collection. With an OLS model, a censored dependent variable can create an econometric inconsistency concern (Greene 2012). To address this, we use two alternative strategies. First, instead of using a continuous form of alive days, we use a binary form of this variable. Specifically, we transform the variable from the number of days on the platform to whether an individual stays on the

¹⁶ To address a potential concern that virtual races attract user segments that are different from users before the introduction of virtual races, we compare individuals who joined the platform before December 2017, when the virtual race was first introduced, with individuals who joined after December 2017. Our observation window spans four years, from June 2015 to May 2019. The first 14-day running distances for the pre-introduction users and post-introduction users are 27.2 km and 27.7 km, respectively, with a statistically insignificant difference ($p > 0.1$). Thus, our results in Section 4 are not driven by new types of users being attracted to virtual races. In addition, we depict the running distance of the first 14 days of the full population before and after the introduction of the virtual race in Figure A2 (Online Appendix). We do not observe a sharp difference after introducing the virtual race feature, indicating that this feature does not substantially change the population attracted to this platform.

¹⁷ The results are similar if we control for runners' total running distance before participating in a virtual race.

platform for more than a period of time, i.e., three months (90 days)¹⁸. Because the last race in our data set finished in May 2019, we collected engagement data on August 25th, 2019, which roughly spans three months. We apply a Probit model to examine the impact of a virtual race as follows:

$$\begin{aligned} \text{Alive3M}_i^* &= \alpha + \beta_1 \times \text{VirtualRace}_i + \Sigma \gamma_w \times \mathbf{Z}_i + \eta_t + \varepsilon_i \\ \text{Alive3M}_i &= 1 [\text{Alive3M}_i^* > 0] \end{aligned} \quad (2)$$

where Alive3M_i^* is the latent variable, and Alive3M_i means individual i stays on the platform for more than three months, and all the independent variables are the same as specified in Equation (1).

Second, we use a Tobit model to address the censoring problem for consistent estimates (Greene 2012). The model is the same as Equation (1), but the latent variable has the following observation rule:

$$\text{AliveDays}_i = \begin{cases} \text{AliveDays}_i, & \text{if } \text{AliveDays}_i \leq 90 \\ 90, & \text{if } \text{AliveDays}_i > 90 \end{cases} \quad (3)$$

3.4 Addressing Endogeneity Concerns

Since participating in a virtual race is a self-selected decision by a runner, it might be affected by several factors, either observable or unobservable. For observables, one deciding factor in participating in a virtual race is users' running capabilities. Thus, there is a need to account for runners' capabilities before reaching the conclusion that participating in a virtual race can be beneficial for new users' longer-term engagement. In addition, some unobserved variables might drive both virtual race participation and platform engagement, which would lead to an upward bias in our estimation of the effect of participating in virtual races. We employ the following three strategies to address these endogeneity concerns: Propensity Score Matching (PSM), Endogenous Treatment Effect Model, and Regression Discontinuity Design, which we elaborate on in more detail below.

3.4.1 Matched Sample Analysis

To account for factors such as individual demographics that simultaneously affect runners' participation in virtual races and long-term engagement, we employ propensity score matching (PSM) to

¹⁸ Other time windows, such as 60 or 180 days, yield similar results.

match observations with and without participating in a virtual race (for further details on propensity score approach, see Rosenbaum and Rubin 1983). PSM can ensure comparability between the treated and control groups. The first stage is a Probit model that uses all observable variables, including gender, age, weight, height, the number of new social connections (online friends) before the virtual race, and running performance (distance) before the virtual race. After calculating the propensity score, we perform a one-to-one nearest-neighbor matching without replacement and then identify observations on a common support region. After identifying a matched sample, we then rerun Equations (1) to (3) to examine the impact of virtual races on user engagement.

3.4.2 Endogenous Treatment Regression Model

To effectively address concerns regarding omitted variables that may simultaneously influence participation in virtual races and long-term engagement, we use the endogenous treatment regression model (Wooldridge 2010), whose estimation is based on Heckman's selection model (Heckman 1978). We apply the treatment effect model described in Barnow et al. (1980) and Cameron and Trivedi (2005, sec. 16.7 and 25.3.4). The procedure has two stages: the first stage is a Probit model used to estimate a reduced form of the virtual race selection equation, and the second stage is an OLS regression to estimate the impact of virtual race on alive days. We follow Pabilonia and Zoghi (2005) and use maximum-likelihood estimation with specifications as below:

$$VirtualRace_i = \begin{cases} 1, & \text{if } \delta W_i + \Sigma \gamma_w \times \mathbf{Z}_i + \eta_t + u_i > 0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$AliveDays_i = \alpha + \beta_1 \times VirtualRace_i + \Sigma \gamma_w \times \mathbf{Z}_i + \eta_t + \epsilon_i \quad (5)$$

where *AliveDays*, *VirtualRace*, and $\mathbf{Z}$ are specified in Equation (1). W_i is the instrumental variable used to model the choice of participation, and the error terms ϵ_i and u_i follow a bivariate normal distribution¹⁹.

¹⁹ According to Kang et al. (2016), a valid instrument needs to satisfy the following three conditions: “(1) related to the exposure (2) have no direct pathway to the outcome and (3) are not related to unmeasured variables that affect the exposure and the outcome (p. 132).”

We leverage a unique feature from the platform to construct instruments. As described previously, the platform uses astrological zodiac signs as the themes for its virtual races: for instance, on March 17th, 2018, the platform held the “Pisces Virtual Race,” and on June 16th, 2018, the platform held the “Gemini Virtual Race.” Since astrology culture is very influential in Taiwan, if a new platform user’s zodiac sign matches the theme of the virtual race, her intention to participate in the race would be higher than those whose zodiac signs do not match the race theme²⁰. Accordingly, we construct an instrumental variable *ZodiacMatch*, to indicate whether the user’s zodiac is the same as the nearest virtual race’s theme. For example, if there are two users registering on March 1st, 2018, the closest virtual race a user can attend is the “Pisces Virtual Race.” Both users have the freedom to choose whether they want to participate in this upcoming virtual race or not. Since anyone born between February 20th and March 20th would be categorized as Pisces, if User1’s zodiac is Pisces (e.g., born on March 10th, 1995), and User2’s zodiac is Scorpio (e.g., born on November 10th, 1991), User1’s *ZodiacMatch* would be 1 and User2’s *ZodiacMatch* would be 0. Since we model a user’s likelihood to participate in the nearest virtual race after registration, this instrument is valid because the significance of astrology in Taiwan would increase the propensity to participate in a particular virtual race if *ZodiacMatch* equals to 1, but the zodiac sign should be unrelated to a user’s engagement in a platform as the platform is zodiac neutral along time. We discuss the validity of the instrument in Section 4.2.2 in greater detail.

3.4.3 Analysis using Fuzzy Regression Discontinuity Design (RDD)

To further corroborate our empirical findings, we exploit the proximity of users’ birthdays as the cutoff and apply the fuzzy regression discontinuity design (RDD)²¹. Fuzzy RDD is appropriate in our context because the probability of the treatment is discontinuous at the cutoff (users whose birthday falls into the zodiac theme of a virtual race), but not to the degree of a definite 0 or 1. We use birthday as the

²⁰ Blogs about virtual races on the platform corroborate this expectation: some specifically mention that “I must attend this race because this is my horoscope.” (<https://medium.com/@pohsuan711/%E6%96%B0%E7%AB%B9-running-2018-%E5%B7%A8%E8%9F%B9%E7%B7%9A%E4%B8%8A%E9%A6%AC%E6%8B%89%E6%9D%BE-2f43c841eda7>)

²¹ We thank an anonymous reviewer and the Associate Editor for the suggestion to apply fuzzy RDD.

running variable. Since a zodiac contains a start date and an end date, the running variable allows an individual to have two directions to fall into the “assignment group.” To make this setting fit with a conventional RDD framework, we calculate the number of days difference between the birthday and starting date or ending date for each user and keep the relatively smaller one. In other words, for each user, we can calculate the minimum distance from their birthdays to the starting date or ending date of the zodiac for the first virtual race after a user registered. The following examples illustrate how we code the running variable. If user A1’s birthday is three days before the start date of the zodiac sign, her running variable would be -3. Likewise, if user A2’s birthday is three days after the end day of the zodiac sign, her running variable would also be -3. For treated users, the calculation is similar. If user B1’s birthday is five days after the zodiac start date, her running variable would be five. B2’s running variable would also be 5 if B2’s birthday is 5 days before the zodiac end day (and 25 days after the starting day if this zodiac sign covers 30 days) of the zodiac sign. Therefore, the running variable spans the range between -168 and 15, with 0 as the cutoff point to separate the control and treatment groups.

To select bandwidth for the RDD method, the commonly used data-driven bandwidth selection method is the algorithm by Imbens and Kalyanaraman (2012) and further improved ones (Calonico et al. 2014). However, while these methods impose bandwidths of the same length for the two sides of the cutoff point, they are not the most applicable ones to our RDD context due to the significant difference between control and treatment groups regarding the density of the running variable (birthday) of the two sides. As discussed in Arai and Ichimura (2018), using a single bandwidth in our context could be problematic – if the bandwidth is big, it will incur a large bias in the treatment group; if the bandwidth is set too small, though the bias could be reduced, a large variance could be generated. To accommodate unique contexts such as ours, Arai and Ichimura (2018) and other economists (Jales et al. 2017) further developed data-driven bandwidth selection algorithms based on the foundation of Imbens and Kalyanaraman (2012) to allow the selection of different bandwidths on both sides of the cutoff points. Specifically, we apply the most suitable bandwidth selection algorithm (Calonico et al. 2017; Calonico et al. 2014) to derive the algorithmically-selected bandwidths. The algorithm applies the mean squared error (MSE) optimal

bandwidth selectors with different optimal bandwidths on both sides of the cutoff points to derive the bandwidths, which result in 62 days below and 8 days above the threshold.

We then apply the fuzzy RDD by using the hazard (survival) model. Survival analysis is especially appropriate in our context because the model does not rely on the normality assumption and provides an approach to address the incomplete observation of survival times when censoring occurs (Hosmer et al. 2008). Previous literature (Martínez-Cambor et al. 2019; Ying et al. 2019) provides a clear procedure to apply the two-stage residual inclusion (2SRI) algorithm (Terza et al. 2008) in non-linear contexts. The 2SRI computes the expected exposure and includes the residuals from the first stage in the second stage. The procedure to apply 2SRI in the hazard model is straightforward. We performed the first step of IV analysis by applying the logistic regression of treatment on the IV. We then calculate the residuals. In the second stage, we included the residuals to fit the survival model. Following Martínez-Cambor et al. (2019), we specify our model below.

$$\text{First-stage:} \quad T_i = \alpha_0 + \delta_1 D_i + \delta_2 R_i + \Sigma \delta_w \times \mathbf{Z}_i + \varepsilon_i \quad (6)$$

$$\text{Second-stage:} \quad h_i(t|\mathbf{X}_i) = h_0(t) * \exp(\beta_1 T_i + \beta_2 R_i + \beta_3 \text{Residual}_i + \Sigma \beta_w \times \mathbf{Z}_i) \quad (7)$$

where $h_i(t|\mathbf{X}_{it})$ is the conditional instantaneous hazard rate of leaving the platform for individual i at time t ²². We define failure for individuals who do not stay for more than 90 days²³. $h_0(t)$ is the baseline hazard function. $T_i = 1$ if individual i participates in a virtual race, and 0 otherwise; $D_i = 1$ if individual i is assigned to the treatment that is above the cutoff point, and 0 otherwise. R_i is the running variable, which is calculated as the minimum distance from a user's birthday to the starting date or ending date of the zodiac for the first virtual race after a user registered. Residual_i is the residual calculated from the first stage. $\mathbf{Z}$ is a vector that includes individuals' demographics, including gender, weight, height, and age.

4. Results: Effects of a Virtual Race on New Users' Platform Engagement

4.1. Main Findings

²² We utilized a Cox regression model to estimate the effect in our analysis. In order to ensure the validity of the model, we evaluated the proportional hazards assumption (Grambsch and Therneau 1994) by conducting the Schoenfeld residual test and confirmed that our model satisfies the proportional hazards assumption.

²³ The results are robust by using other time windows, such as 60 days or 120 days.

Before we discuss the main results, an important consideration we address is to ensure the observed results are driven by participation in virtual races, and that users in the treated and control groups are registered organically. Thus, we examine users' pre-first-race date of registration between participants and non-participants, and show that the treated and control groups are indeed very similar in terms of their timing of registrations before the start of the first virtual race. On average, control users registered 44.5 days before the start date of the first virtual race one can attend and treated users registered 44.4 days before the start date. Thus, our treated users and control users have similar pre-first race registration dates. Figure A3 further reinforces the similarity between the two groups by showing the distributions of the days from the registration date for the first virtual race start date between participants and non-participants.

The main results of a virtual race on new user engagement are presented in Table 4. Column (1) shows that individuals who run the virtual race will stay on the platform 63.9 days longer than individuals who do not participate. In column (2), we use a Probit model to account for the censored outcome variable and examine whether running a virtual race can increase an individual's likelihood of staying more than 90 days. The marginal effect shows that virtual race participants are 16.9% more likely to stay for more than 90 days than non-participants. In column (3), we show that our results are robust when using the Tobit model (to account for right censoring): individuals who run the virtual race will stay on the platform for 39.4 more days than non-participants.

4.2. Results of Analyses Addressing Endogeneity Concerns

4.2.1 Propensity Score Matching

To address the differences of observable factors in the pre-treatment period, we employed a PSM method to ensure the balance of individuals' demographics, such as gender, age, weight, height, and the number of social connections (online friends). Most importantly, we also include the running distance before the virtual race to ensure individuals' running capabilities before virtual races are similar.

We first ensure that all observable variables have no significant difference after matching (see Table A1 for more detail). The results by using the matched sample in Table 5 show that our findings are robust: participating in a virtual race helps new users engage longer in this platform.

4.2.2. Instrumental Variable Estimates

We begin with model-free evidence of the influence of the zodiac on the first stage decision. Individuals who have the same horoscope as the zodiac-based virtual race have a likelihood of 23.8% to participate in the virtual race. For individuals whose astrological sign does not match the race theme, the participation rate is only 6.6%. This provides initial evidence that individuals' own horoscope is correlated with their participation decisions.

Instrumental variable results are reported in Table 6. The first column represents the selection equation, and the second column shows the outcome equation. In the first column, *ZodiacMatch* is significant at $p < 0.01$, indicating that individuals whose zodiac matches the theme of the virtual race are more likely to participate. The second column presents results after correcting for endogeneity; they remain consistent with the main results shown in Table 4: individuals who run a virtual race stay 42.8 days longer than individuals who do not participate in the race (Table 6, column (2)). We conduct Wald tests of ρ , and results show rejection of the null hypothesis, which indicates no correlation between the errors in the selection equation and those in the outcome equation. This shows that *VirtualRace* does suffer from self-selection concerns, providing a rationale for the application of the endogenous treatment regression model²⁴.

To further address the potential right-censoring of the dependent variable, we conduct an instrumental-variable (IV) Tobit model to estimate the effect of participating in a virtual race on the likelihood of survival. Similar to the main results section, we set up the right censor at 90 days (other censoring numbers are also robust). Table A2 shows the results. *ZodiacMatch* in the first column shows support for our argument that individuals with the same zodiac as the closest virtual race's theme have a higher likelihood of participating in the race at $p < 0.01$. Column (2) shows that individuals who participate in a virtual race stay longer by 24.2 days than non-participants. The results, again, indicate that our findings are robust to the censoring issue. In addition, the first-stage F-statistic is 78.4, which is well above the

²⁴ The sample size is slightly smaller than the original sample because not everyone provides the full birthdays when they register. However, the likelihood of participating in a virtual race in this sample is similar to the original sample, alleviating the concerns that this sample might not be representative enough.

commonly used Stock-Yogo critical value of 10, indicating that this IV is not weak. Overall, our instrumental variable approaches (endogenous treatment regression, and IV Tobit model) jointly provide strong support that participating in a virtual race is associated with user engagement with the platform.

To provide further robustness, we show that it is unlikely that individuals' engagement on this platform is directly influenced by their zodiac sign. If the platform provided birthday vouchers or other gifts to individuals as is commonplace on many merchant websites, then individuals' zodiac might not be an appropriate instrumental variable. We confirm with platform managers that *RunningPlatform* does not offer promotional birthday coupons that might affect user engagement. Moreover, the platform does not actively mail any specific items to encourage the participation of virtual races because of cost concerns. In other words, the participation is most likely organic.

In addition, one might be concerned that some individuals join the platform because of the attraction of virtual races, especially when the zodiac sign is matched. If this happens, our current results would represent a conservative estimate because individuals who are attracted by virtual races would likely stop engagement on the platform soon after participating in a virtual race. To provide further robustness of our instrumental variable results, we conduct two additional analyses in Table A3, columns (1) and (2). Platform managers indicated that they do not spend extra marketing resources to promote virtual races. Therefore, there are only two major ways an individual would know about virtual races before joining the platform. The first is through the Facebook fan page of this platform. We exclude individuals who register via a Facebook account. When registering, a user can choose to use a Facebook account to efficiently sign up on the platform, or manually fill in the account information. Users who register via their Facebook account have a higher likelihood of joining the Facebook fan page than non-users. The IV estimation results in Table A3, column (1) are consistent with the main findings even when we exclude users who register via a Facebook account.

The second channel through which individuals can learn about virtual races is through friends' referrals. We create a subsample by retaining only individuals who have no friends when they first join the platform. If a new user is referred by a friend, it is more likely that this user would follow or be followed

by other existing users (who are the new user's friends). The results in Table A3, column (2), again, indicate consistency with the main results. To summarize, our instrumental variable supports the existence of a positive relationship between participating in a virtual race and user engagement.

In addition, we conducted analyses to compare newly registered users before and after the platform launched virtual races (December 2017). If virtual races attract users whose birthdays are close to game day, we should expect to see different distributions after the platform launched this service. To simplify this task, we construct a variable, *SameMonth*, which is 1 if a user's birthday month equals the registered month. We keep our observational window from June 2015 to May 2019. We then calculate the ratio of *SameMonth* for every month before and after the platform introduced virtual race. The ratios between the pre-introduction users and post-introduction users are 8.8% and 8.9%, respectively, and are statistically insignificant ($p > 0.1$). The results show that user registration on this platform is not affected by the virtual race's zodiac theme.

4.2.3 Fuzzy RDD Results

To apply RDD, the first assumption is that there should be a discontinuous jump in the probability of participating in virtual races as the running variable crosses the cutoff point. Figure 2 visually shows support for this assumption. The x-axis represents how far a user's birthday is from the starting/ending date of the zodiac sign, and the y-axis represents the average participation rate for users having the same distance from the starting/ending date of the zodiac sign. We can observe that the participation rates mostly fall below 10% before the cutoff point, and the participation rates jump mostly above 20% above the cutoff value. Another important assumption of the RDD is that the running variable and other covariates should be locally continuous across the threshold. In other words, runners before and after the cutoff points within the bandwidth should not be systematically different. We test the imbalance of available covariates between treated and untreated users within the bandwidth. We find these two groups of users' covariates are balanced: female (47.3% control, 46.7% treated), weight (64.4 vs. 64.2), height (169.4 vs. 169.3), age (35.7 vs. 35.4), number of connections before treatment (10.7 vs. 10.2), and average running distance per run before treatment (5.3 km vs. 5.4 km). All differences are statistically insignificant.

The results of RDD design by applying the hazard model are presented in Table 7. We start with a parsimonious model without controlling covariates in column (1), and then apply the covariates into the model in column (2). The survival model is used to examine users' likelihood of leaving the platform. The results in both columns represent the hazard ratio (exponentiated coefficients), which means that the likelihood of leaving the platform within 90 days is reduced by roughly half in the treatment group (virtual race participants), compared to the control group. Thus, the results in both columns lend strong support for the conclusion that participating in a virtual race can make users engage longer in this platform.

4.3 Robustness Checks

We conduct several additional tests to examine the robustness of our findings. First, our model only examines whether individuals choose to participate in a virtual race that is closest to their registration date but does not consider attendance in subsequent virtual races. Our results, thus, potentially underestimate the effects of virtual races because individuals in the control group might participate in the future. To create a pure nonparticipation control group, we retain only individuals who have never participated in any virtual race on the platform. We present the results in Table A3, column (3). Participating in a virtual race allows users to stay 71.5 more days in this platform than non-participants do, and the effect indeed is higher than the main findings, supporting the claim that our main results underestimate the effects of virtual races.

Second, our main model examined whether participating in “a virtual race that is closest to a user's registration date” would affect user engagement or not. To provide practical insight, we conduct a more conservative analysis by examining the impact of “one” virtual race on user engagement. Specifically, we retain a subsample of users who either have never attended a virtual race (control group) or attended “only one” virtual race regardless of whether the virtual race is closest to a user's registration date or not (treatment group). The results in Table A3, column (4) remain to show that users participating in a virtual race stay longer in this platform.

Third, to further enhance the rigor of our findings, we retain users who eventually attend future races in the sample (regardless of users in the control or treatment groups) to account for intent to generally

run a virtual race. This alludes to some sort of forward matching. Table A3, column (5) shows that the results are qualitatively similar to our main findings.

Fourth, although the last log-in date is a common measure to quantify user engagement with a platform (Rossmann et al. 2016; Viljanen et al. 2016), one might argue that physical activities are more pertinent to the fitness platform's mission and, therefore, better reflect meaningful user engagement. We thus construct an alternative dependent variable using the individual's last exercise date and apply the same instrumental variable model discussed previously. For simplicity, we suppress the first-stage selection model. Table A4, column (1) shows the results. Similar to the main findings, with this alternative outcome measure, we find that participating in a virtual race positively affects users' length of stay in this platform for exercise. Finally, one might be concerned that the length of stay on this platform does not reflect the intensity of user activities. To address this, we use two alternative dependent variables for assessing the intensity of engagement: (1) the number of times a user uploaded running records; (2) the number of times a user logged-in to the platform. We use log transformations of both variables. Results of the instrumental variable estimation are in Table A4, columns (2) and (3). Individuals who participate in a virtual race upload 24.2% ($e^{0.217}-1$) more running records and log-in 27.9% ($e^{0.246}-1$) more than non-participants. This indicates that individuals who stay longer indeed exercise more and engage more in this platform, alleviating the concerns that length of stay might not be adequate as a measure of user engagement²⁵.

4.4. Exploring Underlying Mechanisms

4.4.1 Mechanisms Motivating User Engagement

Our results demonstrate the effectiveness of virtual races but do not reveal *why*. To explore underlying mechanisms, we focus on one key difference between online and offline racing — the form of social interactions. Compared with traditional offline races, although a virtual race cannot support direct social interactions physically, it greatly reduces the barriers to join a race as well as efficiently connects

²⁵ In order to examine whether the results hold after the initial increased running activity in preparation for the virtual race (and during the virtual race), we exclude data before the end of virtual races and rerun the model. The results remain to indicate that users who participate in the first virtual race are more likely to increase their number of running records and the number of log-ins.

one runner to a broader base of runners by overcoming geographical and temporal limitations. In fact, through a combination of mobile technology and social networking, a virtual race might offer unique forms of social interactions that are not available in offline races. In offline races, participants may interact extensively with a small number of familiar peers (or friends). However, information about the vast majority of other runners is not easily accessible, making it difficult to find matched peers or new friends. In contrast, a virtual race releases the location constraint. By fulfilling social needs in a more efficient and expansive manner than its offline equivalent, a virtual race can facilitate social interactions that complement running experiences and benefit runners.

We attempt to verify this mechanism by focusing on heterogeneity among runners arising from their cost of social interactions and social needs. To address the practical challenge of identifying which individuals incur greater costs or which individuals have a stronger desire to seek social support, we conduct a variety of analyses. First, we use a proxy to differentiate runners' social costs based on registration from a location outside Taiwan (i.e., non-local registration)²⁶. Since the platform is largely used by runners from Taiwan²⁷, we make a reasonable assumption that individuals who live outside Taiwan incur higher costs in fulfilling social needs. In our sample, roughly 7% of users registered outside Taiwan. Results are presented in Table 8. Column (1) shows a positive and significant interaction term, indicating that virtual races have a significant impact on the engagement of users who register outside Taiwan. We also see that the effect of a virtual race is stronger for individuals who register from other locations than it is for individuals in Taiwan. These results indicate that virtual races provide a forum to satisfy social needs for those who experience higher costs associated with in-person social interactions. In addition to using individuals' registration location as an indicator, we also use individuals' GPS to ensure whether they exercise outside Taiwan as a robustness check. For individuals who have uploaded running records, 5.65% of users always upload their running records outside Taiwan. The results (omitted for brevity) are broadly similar to the results in Table

²⁶ Users must specify where they live when they sign up for the platform.

²⁷ Conversations with the platform manager indicate that most people are Taiwanese even if they registered outside Taiwan, and the majority of them use traditional Chinese as the language for interaction.

8, column (1), supporting our argument that people who register (and keep using this app) outside Taiwan benefit more from the virtual races.

Next, we use a proxy to differentiate runners based on social needs measured by their prior social interactions on the platform. *RunningPlatform* allows users to form running clubs of varying sizes. Each club represents a “social circle,”: users in the online running club can see each other’s physical activity, chat with other runners, or compare the ranks of all users in a club. We make a reasonable assumption that individuals who participate in running clubs are individuals who have a stronger desire to seek social interactions. Therefore, the benefits of virtual races might be stronger for this group of users. Results, in Table 8, column (2), show that runners who participated in a running club before the first virtual race benefit more from the virtual race than others. Those runners who are members of running clubs stay 14.7 days longer, supporting our argument that virtual races can be especially useful for those who have social needs²⁸.

Moreover, we argued that virtual races are distinct from other social features such as leaderboards because virtual races can continue to be effective even when users do not have many social connections. We empirically examine this argument by keeping users who do not have any online friends before participating in virtual races. The results are presented in Table A5, and support our assertion that virtual races increase engagement even for runners who have no social connections at all. The results have practical implications for executives: virtual races may be especially useful for new users because such races, while leveraging the community’s social features, do not need high social connections to be effective.

Finally, to further verify the above mechanism, we examine users’ daily new social connections to investigate whether virtual races provide additional social connections that could yield an increase in engagement. We calculate individuals’ daily new connections (follower + followed). The platform provides directional social network features; therefore, a user can follow or be followed by other users. To estimate the impact of a virtual race on individuals’ social connections, we employ a difference-in-differences model. To control for pre-treatment differences, we used the aforementioned matched sample (see Section 3.4.1).

²⁸ The results are qualitatively similar if we apply the IV models discussed in the previous section.

We then compare new social connections between the treatment group and control group in the pre-treatment and the post-treatment periods with the following specification:

$$Y_{it} = \beta_0 + \beta_1 \times VirtualRace_i + \beta_2 \times Post_t + \beta_3 \times VirtualRace_i \times Post_t + \Sigma \gamma_w \times Z_{it} + \eta_t + \varepsilon_{it} \quad (8)$$

where i indexes individual runners and t indexes the period. Y_{it} represents daily new social connections (# follower + # followed). *VirtualRace* represents whether a runner participates in a virtual race or not. $Post = 1$ if t is after the virtual race start date; 0, otherwise. Z is a vector of control variables, including age, gender, weight, height. η_t is the virtual race fixed effect that controls for time-variant unobserved factors affecting individuals' behaviors.

Estimation results are presented in Table 9²⁹. In column (1), we use a 21-day window to examine the effect of virtual racing on individuals' daily new connections. The interaction term is positive and significant at the $p < 0.1$ level. Considering the average number of new connections per day during this 42-day (21 days in the pre-period and 21 days in the post-period) window is 0.014, a 0.006 increase represents a 42.9% increase in social connections. This result indicates that activities involving robust social interactions can continue to be effective when they move online. In addition to using a 21-day window, we test a 9-day window because participants need to complete the race within 9 days after the start date. The results, presented in Table 9, column (2), are qualitatively similar. We again find that virtual races can be beneficial for new users not only by increasing long-term engagement but also by helping them to form social connections.

²⁹ We conduct a formal test for the parallel trend assumption. Meyer (1995) suggests examining the validity threat by using the data in multiple pre-treatment periods. Specifically, we follow previous work (Kumar and Telang 2012) to test the trend difference in the pre-treatment period. We first divide the pre-treatment period into three periods, which are 3-weeks before the treatment, 2-weeks before treatment, and 1-week before treatment. We then redo the regression analysis only in the pre-treatment period. While all the variables remain the same, the *Post* variable will be replaced by using pre-treatment week dummies. If the parallel trend assumption holds, we expect to see insignificant coefficients for the interaction terms. It means individuals in the treatment group are not different from pre-treatment week 3 to pre-treatment week 1 as compared to individuals in the control group. The results in Table A6 show that both interaction terms are insignificant, indicating that individuals in the treatment group are similar to individuals in the control group in terms of making daily new friends. We thus conclude that the parallel trend assumption is not violated.

A plausible concern may be that social connections drive the increased virtual races sign-up rate and bias our findings. We rule out this possibility in the following analyses. First, we empirically examine whether participating in running clubs causes people to sign up for virtual races or not. The t-test shows that the likelihood to participate in a virtual race for running-club users (8.47%) is statistically indifferent from that of non-running club users (8.01%) ($p > 0.1$). Second, we compare the number of friends before attending virtual races between users who participate in virtual races and those who do not attend virtual races. We find that the numbers of social connections between these two groups of users are also statistically insignificant (11.09 (non-virtual race users) vs. 10.94 (virtual races users); $p > 0.1$). Thus, the existing social connections (joining running clubs or having more friends) do not drive the increased virtual race sign-up rate.

4.4.2 Other Possible Mechanisms Motivating User Engagement

Other than social needs, virtual races may provide alternative mechanisms to motivate greater engagement with the platform. We have argued that a major strength of virtual races, as compared to offline races, is the low entry costs. Individuals who have GPS-enabled devices, such as smartphones or sport watches, which track their run can participate in a virtual race. This provides many beginners (new users) a chance to experience the flavor of the races and the atmosphere of competition and learn more about their own physical capabilities. Thus, it is possible that virtual races could be more beneficial for beginners than sophisticated runners because sophisticated runners tend to have greater self-knowledge about their own abilities. Since users in our sample are all newly registered users, using the running distances might not be ideal because they have not had enough time to accumulate running records. Therefore, we use a more intuitive instrument to allow us to differentiate different types of runners: the running devices. Individuals who use mobile devices are reasonably less sophisticated than individuals who use dedicated wearable devices. For instance, Wiesner et al. (2018) reported that, among users who use technologies, 60% of them use GPS-equipped sports watches; only 24% of them use smartphones and apps during a half-marathon and marathon competition, indicating that most professional runners tend to use wearable devices instead of mobile devices to track their physical activity, likely because of weight and battery constraints. We label

users who mainly use mobile devices to upload their running records as mobile users. In Table 8, column (3), we see that mobile users gain more benefit from participating in a virtual race than those who use wearables, indicating that virtual racing can help this group of new and relatively inexperienced users stay longer on this platform.

4.5. Additional Exploration

We conduct several additional analyses to provide a more comprehensive understanding of the impact of virtual races. First, we explore whether virtual race participation increases a customer's value for the platform³⁰ in terms of spending. *RunningPlatform* offers races for free and generates revenue through its e-commerce sales of sports-related gadgets to users. We collected users' spending in the e-commerce portal and created a new dependent variable to capture the total lifetime spending of users. The results of the endogenous treatment effect model are presented in Table A4, column (4) (the first stage is omitted for brevity). The results show that attending a virtual race can increase individuals' total spending on the e-commerce portal by 48.4% ($= e^{0.395} - 1$)³¹.

Second, we examine whether participating in a virtual race would increase the likelihood of attending future races. The results are presented in Table A7, column (1). The results show that individuals who participate in the first virtual race are more likely to attend the next virtual race.

Third, a virtual race might serve as a self-reference, which is especially important for athletes because it allows for an adjustment of one's own pace in the long run (Ray-Smith et al. 2019). Such assessments could help participants develop better insight and estimates of their capabilities and performance that then allows them to evaluate the feasibility of participating in an offline race more

³⁰ We thank the anonymous reviewer for suggesting this alternative dependent variable.

³¹ A plausible concern with this analysis is whether the effect is driven by virtual race participants buying medals after the race and not buying anything else on the platform. Due to data limitations, we use a simple strategy to address this concern. Specifically, we find participants attending the first virtual race would increase their purchases by 332.5 New Taiwan Dollars (NTD) (558.8 NTD of treated users vs. 226.3 NTD of control users). In our sample, roughly 39% of participants purchase a medal at a cost of 99 NTD; thus, the expected spending on medals is 38.6 (99×0.39) NTD. We see that participation in the first virtual race can still increase revenue by roughly 293.9 ($332.5 - 38.6$) NTD for the focal platform. Even if we use the most conservative way and assume every participant purchased a medal, participation in the first virtual race can still increase revenue of non-medal products by 233.5 ($332.5 - 99$).

accurately. In addition, a virtual race facilitates a sense of competition but with low-pressure and a sense of accomplishment among participants, which could ultimately motivate individuals to try offline races.

On this platform, any individual who participates in an offline race can indicate their participation or report their running records. To examine our conjecture, we collect all the offline race records until the last day of our data collection (August 2019). Using this, we are able to quantify whether an individual participates in offline races or not after participating in the first virtual race. We apply the endogenous treatment effect model to address endogenous concerns (the first stage is omitted for brevity). The results are presented in Table A8, column (1). The results show that virtual races have a positive and significant effect on individuals' likelihood of participating in offline races. Since this information is self-reported, we downloaded a list of names for individuals who participate in official races in Taiwan³² to address this concern. We compare the platform user's name with the official name used in the offline race by using exact matching (i.e., the names need to be identical), and check whether a platform user attends any of the official offline marathon races after the end date of their closest virtual race. For rigor, we keep users who use Chinese names on the platform because most runners in official, offline races use their own formal Chinese names. We use the same instrumental variable approach to address endogeneity concerns. We then check whether a user attends any of the offline races after the end of virtual races. The results are presented in Table A8, column (2). Similar to previous results, participation in a virtual race is positively associated with an increased likelihood of attending offline races³³.

5. The Impact of a Virtual Race on Physical Activities

5.1 Data and Empirical Models

Our second research question asks if a virtual race affects physical activity. Our previous sample is not suitable to answer this question because (i) most users are newly registered users that prevents us

³² Chinese Taipei Road Running Association. <https://www.sportsnet.org.tw/score.php>

³³ We acknowledge that causality for the outcome of participating in offline virtual races cannot be established without more granular data. Thus, we treat the results more as an association and are conservative about inferring a causal relationship.

from comparing their physical activity before and after participating in a virtual race; (ii) physical activity would be naturally affected by individuals' tenure, which is also affected by virtual races.

Therefore, to understand the impact of participation in a virtual race on physical activity, we construct a new sample, *Sample 2*. We collected all existing users' physical activity three weeks before and three weeks after the same 18 virtual races used in *Sample 1*³⁴. For each virtual race, we can then separate users into a treated group (participants) and a control group (non-participants). Collecting six weeks of data allows us to employ a difference-in-differences (DID) model and observe the change in users' physical activity before and after treatment (virtual race participation). Therefore, unlike *Sample 1* that uses new users, *Sample 2* consists of existing users to ensure that running data is sufficient and stable. To enable a fair comparison, we use propensity score matching (PSM) to create a balanced sample. Table 2-2 shows the summary statistics, and Table 3-2 shows correlations. On average, individuals run roughly 17 km per week³⁵. With 1 to 1 matching between the treatment and control group, the proportion of individuals participating in a virtual race is 50%. Roughly 30% of users are female, and the average age of participants is 39 years. The final sample contains 222,288 user-week observations, which include 37,048 users in both treatment and control groups.

To estimate the impact of a virtual race on individuals' physical activity, we compare the running distance between the treatment group and control group in the pre-treatment and the post-treatment periods with the following DID specification:

$$Y_{it} = \beta_0 + \beta_1 \times VirtualRace_i + \Sigma \beta_m \times Week_t + \Sigma \beta_m \times VirtualRace_i \times Week_t + \Sigma \gamma_w \times Z_{it} + \eta_t + \varepsilon_{it} \quad (9)$$

where i indexes individual runners and t indexes the period (e.g., one week before treatment or two weeks after treatment). Y_{it} represents running distance in a week. $VirtualRace$ represents whether a runner participates in a virtual race or not. $Week = [-3 \text{ to } 2]$ represents the week before or after the treatment. $Week$

³⁴ We use a three-week window to avoid multiple treatment confounds because the gap between each virtual race ranges from three to four weeks. Moreover, since we observe the changes in physical activity before and after treatment, we remove individuals who do not have pre-treatment running records.

³⁵ The running distance has been winsorized by the top and bottom one percent.

equal to -1 is the baseline group, which will be omitted in the model, and *Week* greater or equal to 0 represents the week in the treatment period. $\mathbf{Z}$ represents a vector of control variables, including age, gender, weight, and height. η_t is the virtual race fixed effects.

For the DID specification, we need to ensure the difference between the treatment and control groups is minimal, especially the pre-treatment performance (running distances). We use propensity score matching to balance the treatment and control groups and ensure these two groups are comparable (Rosenbaum and Rubin 1983). We follow the standard matching procedure by using observable factors such as gender, age, weight, height, and running distance in the pre-treatment period. Table A9 in the Appendix shows the univariate tests before and after treatment. All the characteristics, including the pre-treatment distance, show no significant differences.

5.2 How Virtual Races Affect Runners' Physical Activities

Our analysis requires that the parallel assumption is not violated, i.e., pre-treatment trends between the treatment and control groups are similar. To formally test whether there is a difference in the pre-treatment period between treatment and control groups, Meyer (1995) suggests examining the validity threat by using data in multiple pre-treatment periods. Therefore, we include multiple pre-treatment periods in our model. Table 10, column (1) presents the results. We use one week before the treatment as the baseline group. We then interact the week dummy variables with the main independent variable, *VirtualRace*, indicating whether an individual runner participates in a virtual race or not. The interaction term in the pre-treatment week shows no statistical difference; this suggests that there is no evidence showing our parallel trend assumption is violated.

The interaction terms in the post-treatment period in column (1) Table 10 show a statistically significant difference. The coefficient $VirtualRace \times 0 \text{ week since VR}$ is 2.82 at $p < 0.01$ level, indicating that individuals in the treatment group would run an additional 2.82 km between 0 and 6 days after the start of virtual race compared to individuals in the control group. The effects gradually reduce to 0.71 km between 7 and 13 days (week 2) and 0.63 km between 14 and 20 days (week 3) after treatment, but they remain statistically significant.

To quantify the impact of a virtual race on individuals' overall physical activity, we aggregate the running distances three weeks before and after the treatment and run a difference-in-difference model. The results in Table 10, column (2) show that the treatment group outruns the control group in the post-treatment period by 4.41 km. With a mean of 50.62 km in the pre-treatment period, this represents an 8.7% increase in performance³⁶. Finally, similar to the previous sample, we examine whether participating in a virtual race would increase the likelihood of attending the next virtual race in *Sample 2*. We present the results in Table A7, column (2). We find that existing users are also more likely to run for the next virtual race if they have already participated in one.

6. Conclusion

6.1 Main Results

Connected fitness is an important application domain in the ongoing trend of digitizing social activities that used to occur offline. At the core of this trend is the digital platform that enables user interactions. Despite the debate of whether the highly experiential component of communal physical activities can be successfully virtualized and attract new users, empirical insights have been lacking. Using a running platform as our research setting, we examine how a virtual race can affect new user engagement on the platform. By matching the theme of the virtual races to an individual's own astrological sign, we created an instrumental variable to examine the relationship between virtual races and user engagement with the platform. We find that new users who participate in a virtual race that is closest to their registration date increase their engagement with the platform. We applied an extensive battery of analyses, including PSM, endogenous treatment effect model, fuzzy regression discontinuity design, and a series of robustness checks to support the positive relationship between participation in a virtual race and new user engagement.

At a mechanism level, we unveil novel findings related to the heterogeneous effects of virtual races and who benefits more. Results show that a virtual race can be especially advantageous for individuals who

³⁶ Likewise, our outcomes might suffer from endogenous concerns. We, thus, use the same instrumental variable (*ZodiacMatch*) on the endogenous treatment effect model, and present the results in Table A10. The results are qualitatively similar to Table 10, indicating that our results are robust.

incur higher costs of seeking social interactions (e.g., those who register from a location outside Taiwan) or individuals who have a stronger desire to seek social interactions (e.g., those who participate in a running club). We further show that participation in a virtual race continues to be effective even if users do not have any social connection, supporting our arguments that a virtual race offers a common experience that can increase the emotional bond of users to the platform. In addition, such races help individuals to build online social connections, boosting daily connections by a striking 64%. Moreover, we show that recreational runners (mobile app users) benefit more from participating in a virtual race than sophisticated ones (wearable device users) who are likely competitive. Our mechanism exploration provides strong evidence that a virtual race can be beneficial to new users because it provides social supports, facilitates social connections, and serves users who do not have high social connections.

We conducted additional analyses to establish that individuals increase their spending on this platform by 48.4% if they participated in a virtual race when they first registered. In addition, the experience of participating in a virtual race not only increases their likelihood of attending future virtual races, but also increases their likelihood to participate in offline races. Lastly, by using the difference-in-difference model and existing users as an additional sample, we show that virtual races increase individuals' physical activities by 4.41 km in a three-week period, as compared with other participants who do not join virtual races as the baseline.

6.2 Research Contributions, Practical Implications, and Limitations

Our work is relevant to the literature on digital health in information systems and offers three important contributions for research. First, as we indicated previously, the multi-billion dollar industry of connected fitness is a result of striking developments in mobile and wearable technologies intersecting with the power of social platforms. While technologies such as mobile apps (Ghose et al. 2021) and wearable devices (Adjerid et al. 2021) have been shown to affect health outcomes, few studies have examined the effect of virtualization of traditionally offline communal fitness events on individuals' physical activities and engagement. In this work, we contribute to the field by providing evidence that, if designed properly, virtual races can be an effective means of connected fitness that increases the chance to expose traditional

offline events (races) via digital technologies to broader audiences because of scalability and accessibility advantages.

For the healthcare field, our study contributes by showing that a virtual race has the potential to help new users form healthy behaviors or to engage new users with exercise platforms, which is inherently an extremely difficult challenge for effortful physician activity. Participating in a virtual race can serve as an effective strategy for new users to overcome the initial resistance to exercise and help them cross the threshold of habit formation related to physical activities. Our findings are especially important considering that one of the pressing challenges for helping individuals increase physical activity is not in motivating them to start running, but rather, the sustainability of such behaviors (Krebs and Duncan 2015). Given the low-cost nature of virtual events, policymakers might consider encouraging or subsidizing such events to enhance the health of the general public. In addition, we find that virtual races can also be useful in increasing existing users' physical activities. Thus, such a virtual event can be viewed as a strategy to improve users' long-term health. Yet, more evidence is needed to support the causal relationship between participating in virtual races and long-term physical activity improvements.

Our work also contributes to the stream of literature examining what elements are needed to make virtual events (e.g., Connected Fitness) more effective. Our study reveals that when the virtualization of fitness events is being considered, enabling social interactions should be a key design feature for the host platform. In our case, the symbolic connection to others that virtual race participation provides can be beneficial for runners who are geographically isolated or who need more social interactions as indicated by their revealed preference in joining a running club. Interestingly, while social interactions and social supports are needed to be effective in virtual races, for a focal user, there is no need to build high social connections, a cold start problem that many social features (such as leaderboards) face. Our results show that virtual races continue to be an effective means to engage users because common experience or social supports need not be derived only from close friends, they can be obtained even from strangers.

Our results also offer practical implications. Virtual races are on the rise. Our study helps reveal that digital platforms combining social networks and mobile tracking features can provide alternative

options for runners to enjoy races, and in turn, foster engagement and physical activities. From the platform managers' perspective, it is imperative to understand whether they should encourage users to participate in a virtual race. Platform user engagement and involvement have long challenged platform managers; both practitioners and researchers are trying hard to incentivize individuals to stay engaged (e.g., Claussen et al. 2013; Gallus 2017). Our results suggest that individuals who participate in a virtual race not only increase their retention rates, but also increase their spending on the e-commerce platform. Managers could consider providing some forms of incentives to encourage newcomers to join the first virtual race because it helps users increase engagement and can potentially increase their social connections, ultimately yielding long-term benefits for the platform.

We acknowledge the limitations of our research. The platform offers a light version of virtual races, meaning that individuals can only observe the exercise outcomes and interact with other runners after finishing the races. Some platforms (e.g., the Codoon physical activity app) provide individuals with live interactions with other participants, and 24-hour live streaming of the status of all runners. Future work with more granular data or more advanced capabilities that connect individuals with others might explore how such functions can further improve runners' physical activities and amplify engagement with the platform. We note that our current data only allows us to quantify the effects of virtual races on users' physical activities in a three-week window. Future studies with adequate resources and designs should trace virtual races' effects on a longer time window.

From the dawn of the 21st century, the virtualization of human activities driven by technological advances has continued to accelerate. The first two decades have been primarily focused on activities that are intrinsically informational by nature, with the separation of bits from atoms in online transactions and social media. The connected fitness movement represents a new level of virtualization of inherently physical activities. Our study provides initial evidence of the positive impact on users when a traditionally physical event with a high-touch experience is virtualized. As the trend of virtualization continues, we call for more research to leverage its potential to design innovative solutions to create business value and address long-lasting societal challenges.

References

- Adjerid, I., Loewenstein, G., Purta, R., and Striegel, A. 2021. "Gain-Loss Incentives and Physical Activity: The Role of Choice and Wearable Health Tools," *Management Science* (64:4), pp. 2642–2667.
- Adjerid, I., Purta, R., Striegel, A., and Loewenstein, G. 2018. "Aggressive Economic Incentives and Physical Activity: The Role of Choice and Technology Decision Aids." Retrieved from <https://arxiv.org/abs/1810.06698> on 11/12/2020.
- Andersen, J. J. 2019. "The State of Running 2019," RunRepeat.com and the IAAF.
- Aragon, S. R., and Johnson, E. S. 2008. "Factors Influencing Completion and Noncompletion of Community College Online Courses," *American Journal of Distance Education* (22:3), pp. 146-158.
- Arai, Y., and Ichimura, H. 2018. "Simultaneous Selection of Optimal Bandwidths for the Sharp Regression Discontinuity Estimator," *Quantitative Economics* (9:1), pp. 441-482.
- Association of International Marathons and Distance Races. 2016. "Marathons in History with >30,000 Finishers." from http://aims-worldrunning.org/statistics/World's_Largest_Marathons.html on 02/09/2020
- Barnow, B., Cain, G., and Goldberger, A. 1980. "Issues in the Analysis of Selectivity Bias," *Evaluation Studies Review Annual* (5), pp. 43-59.
- Berkowitz, L. R., Dzara, K., and Simpkin, A. L. 2021. "Building Your “Educational Peloton:” Cycling Together for Success During Uncertain Times," *Journal of Continuing Education in the Health Professions* (41:1), pp. 8-9.
- Bloom, N., Liang, J., Roberts, J., and Ying, Z. J. 2015. "Does Working from Home Work? Evidence from a Chinese Experiment," *The Quarterly Journal of Economics* (130:1), pp. 165-218.
- Bojd, B., Song, X., Tan, Y., and Yan, X. 2022. "Gamified Challenges in Online Weight-Loss Communities," *Information Systems Research* (Articles in Advance).
- Breevaart, K., Bakker, A. B., and Demerouti, E. 2014. "Daily Self-Management and Employee Work Engagement," *Journal of Vocational Behavior* (84:1), pp. 31-38.
- Bunnage, L. A. 2014. "Social Movement Engagement over the Long Haul: Understanding Activist Retention," *Sociology Compass* (8:4), pp. 433-445.
- Calonico, S., Cattaneo, M. D., Farrell, M. H., and Titiunik, R. 2017. "Rdrobust: Software for Regression-Discontinuity Designs," *The Stata Journal* (17:2), pp. 372-404.
- Calonico, S., Cattaneo, M. D., and Titiunik, R. 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs," *Econometrica* (82:6), pp. 2295-2326.
- Cameron, A., and Trivedi, P. 2005. *Microeconometrics: Methods and Applications*. Cambridge U Press.
- Charness, G., and Gneezy, U. 2009. "Incentives to Exercise," *Econometrica* (77:3), pp. 909-931.
- Chen, W., Wei, X., and Zhu, K. 2017. "Engaging Voluntary Contributions in Online Communities: A Hidden Markov Model," *MIS Quarterly* (42:1), pp. 83-100.
- Cheshire, C., and Antin, J. 2008. "The Social Psychological Effects of Feedback on the Production of Internet Information Pools," *Journal of Computer-Mediated Communication* (13:3), pp. 705-727.
- Choudhury, P., Foroughi, C., and Larson, B. 2021. "Work-from-Anywhere: The Productivity Effects of Geographic Flexibility," *Strategic Management Journal* (42:4), pp. 655-683.
- Ciampaglia, G. L., and Taraborelli, D. 2015. "Moodbar: Increasing New User Retention in Wikipedia through Lightweight Socialization," *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*, pp. 734-742.
- Claussen, J., Kretschmer, T., and Mayrhofer, P. 2013. "The Effects of Rewarding User Engagement: The Case of Facebook Apps," *Information Systems Research* (24:1), pp. 186-200.
- Dagger, T. S., and Danaher, P. J. 2014. "Comparing the Effect of Store Remodeling on New and Existing Customers," *Journal of Marketing* (78:3), pp. 62-80.
- DellaVigna, S., and Malmendier, U. 2006. "Paying Not to Go to the Gym," *American Economic Review* (96:3), pp. 694-719.

- Edwards, E. A., Lumsden, J., Rivas, C., Steed, L., Edwards, L., Thiyagarajan, A., Sohanpal, R., Caton, H., Griffiths, C., and Munafò, M. 2016. "Gamification for Health Promotion: Systematic Review of Behaviour Change Techniques in Smartphone Apps," *BMJ Open* (6:10), p. e012447.
- Eyler, A. A., Brownson, R. C., Donatelle, R. J., King, A. C., Brown, D., and Sallis, J. F. 1999. "Physical Activity Social Support and Middle-and Older-Aged Minority Women: Results from a Us Survey," *Social Science & Medicine* (49:6), pp. 781-789.
- Frankel, R., and Li, X. 2004. "Characteristics of a Firm's Information Environment and the Information Asymmetry between Insiders and Outsiders," *Journal of Accounting and Economics* (37:2), pp. 229-259.
- Gallus, J. 2017. "Fostering Public Good Contributions with Symbolic Awards: A Large-Scale Natural Field Experiment at Wikipedia," *Management Science* (63:12), pp. 3999-4015.
- Ghose, A., Guo, X., Li, B., and Dang, Y. 2021. "Empowering Patients Using Smart Mobile Health Platforms: Evidence from a Randomized Field Experiment," *MIS Quarterly* (46:1), pp. 151-192.
- Gneezy, U., and Rustichini, A. 2000. "Pay Enough or Don't Pay at All," *The Quarterly Journal of Economics* (115:3), pp. 791-810.
- Grambsch, P. M., and Therneau, T. M. 1994. "Proportional Hazards Tests and Diagnostics Based on Weighted Residuals," *Biometrika* (81:3), pp. 515-526.
- Greene, W. H. 2012. *Econometric Analysis*, (7th ed.). Upper Saddle River, NJ: Prentice Hall.
- Hamine, S., Gerth-Guyette, E., Faulx, D., Green, B. B., and Ginsburg, A. S. 2015. "Impact of Mhealth Chronic Disease Management on Treatment Adherence and Patient Outcomes: A Systematic Review," *Journal of Medical Internet Research* (17:2), p. e3951.
- Hart, C. M., Friedmann, E., and Hill, M. 2018. "Online Course-Taking and Student Outcomes in California Community Colleges," *Education Finance and Policy* (13:1), pp. 42-71.
- Heckman, J. J. 1978. "Dummy Endogenous Variables in a Simultaneous Equation System," *Econometrica* (46:4), pp. 931-961.
- Holder, B. 2007. "An Investigation of Hope, Academics, Environment, and Motivation as Predictors of Persistence in Higher Education Online Programs," *The Internet and Higher Education* (10:4), pp. 245-260.
- Hosmer, D. W., Lemeshow, S., and May, S. 2008. *Applied Survival Analysis: Regression Modeling of Time to Event Data*, (2nd ed.). Wiley.
- Hu, X. 2005. "A Data Mining Approach for Retailing Bank Customer Attrition Analysis," *Applied Intelligence* (22:1), pp. 47-60.
- Huang, K.-Y., Chengalur-Smith, I., and Pinsonneault, A. 2019. "Sharing Is Caring: Social Support Provision and Companionship Activities in Healthcare Virtual Support Communities," *MIS Quarterly* (43:2), pp. 395-424.
- Hydari, M. Z., Adjerd, I., and Striegel, A. 2022. "Health Wearables, Gamification, and Healthful Activity," *Management Science* (Article in Advance).
- Imbens, G., and Kalyanaraman, K. 2012. "Optimal Bandwidth Choice for the Regression Discontinuity Estimator," *The Review of Economic Studies* (79:3), pp. 933-959.
- Jales, H., Ma, J., and Yu, Z. 2017. "Optimal Bandwidth Selection for Local Linear Estimation of Discontinuity in Density," *Economics Letters* (153), pp. 23-27.
- Jamison, J. C., Karlan, D., and Raffler, P. 2013. "Mixed Method Evaluation of a Passive Mhealth Sexual Information Texting Service in Uganda," National Bureau of Economic Research.
- Joyce, E., and Kraut, R. E. 2006. "Predicting Continued Participation in Newsgroups," *Journal of Computer-Mediated Communication* (11:3), pp. 723-747.
- Kang, H., Zhang, A., Cai, T. T., and Small, D. S. 2016. "Instrumental Variables Estimation with Some Invalid Instruments and Its Application to Mendelian Randomization," *Journal of the American Statistical Association* (111:513), pp. 132-144.
- Kato-Lin, Y.-C., Abhishek, V., Downs, J. S., and Padman, R. 2016. "Food for Thought: The Impact of M-Health Enabled Interventions on Eating Behavior," in: *SSRN 2736792*.

- Krebs, P., and Duncan, D. T. 2015. "Health App Use among Us Mobile Phone Owners: A National Survey," *JMIR mHealth and uHealth* (3:4), p. e101.
- Kumar, A., and Telang, R. 2012. "Does the Web Reduce Customer Service Cost? Empirical Evidence from a Call Center," *Information Systems Research* (23:3-part-1), pp. 721-737.
- Laibson, D. 1997. "Golden Eggs and Hyperbolic Discounting," *The Quarterly Journal of Economics* (112:2), pp. 443-477.
- Lister, C., West, J. H., Cannon, B., Sax, T., and Brodegard, D. 2014. "Just a Fad? Gamification in Health and Fitness Apps," *JMIR Serious Games* (2:2), p. e3413.
- Liu, C.-W., Gao, G. G., and Agarwal, R. 2021. "Reciprocity or Self-Interest? Leveraging Digital Social Connections for Healthy Behavior," *MIS Quarterly* (46:1), pp. 261-298.
- Looyestyn, J., Kernot, J., Boshoff, K., Ryan, J., Edney, S., and Maher, C. 2017. "Does Gamification Increase Engagement with Online Programs? A Systematic Review," *PLoS One* (12:3).
- López, O. S. 2013. "Creating a Sustainable University and Community through a Common Experience," *International Journal of Sustainability in Higher Education* (14:3), pp. 291-309.
- Martínez-Camblor, P., Mackenzie, T., Staiger, D. O., Goodney, P. P., and O'Malley, A. J. 2019. "Adjusting for Bias Introduced by Instrumental Variable Estimation in the Cox Proportional Hazards Model," *Biostatistics* (20:1), pp. 80-96.
- Meyer, B. D. 1995. "Natural and Quasi-Experiments in Economics," *Journal of Business & Economic Statistics* (13:2), pp. 151-161.
- Meyers, J. 2017. "As Running Booms in China, Marathoners Look to Boston." *BostonGlobe*, from retrieved from <https://www.bostonglobe.com/business/2017/09/11/running-booms-china-marathoners-look-boston/CyNIIUrjFVkoBet488xPUK/story.html> on 02/09/2020
- Milne-Ives, M., Lam, C., De Cock, C., Van Velthoven, M. H., and Meinert, E. 2020. "Mobile Apps for Health Behavior Change in Physical Activity, Diet, Drug and Alcohol Use, and Mental Health: Systematic Review," *JMIR mHealth and uHealth* (8:3), p. e17046.
- Morgan, J. T., and Halfaker, A. 2018. "Evaluating the Impact of the Wikipedia Teahouse on Newcomer Socialization and Retention," *Proceedings of the 14th International Symposium on Open Collaboration*, pp. 1-7.
- Morris, L. V., Wu, S.-S., and Finnegan, C. L. 2005. "Predicting Retention in Online General Education Courses," *The American Journal of Distance Education* (19:1), pp. 23-36.
- Müller, A. M., Khoo, S., and Morris, T. 2016. "Text Messaging for Exercise Promotion in Older Adults from an Upper-Middle-Income Country: Randomized Controlled Trial," *Journal of Medical Internet Research* (18:1), p. e5.
- Murphy, C. A., and Stewart, J. C. 2017. "On-Campus Students Taking Online Courses: Factors Associated with Unsuccessful Course Completion," *The Internet and Higher Education* (34), pp. 1-9.
- Nawrot, I., and Doucet, A. 2014. "Building Engagement for Mooc Students: Introducing Support for Time Management on Online Learning Platforms," *Proceedings of the 23rd International Conference on World Wide Web*, pp. 1077-1082.
- Nitzan, I., and Libai, B. 2011. "Social Effects on Customer Retention," *Journal of Marketing* (75:6), pp. 24-38.
- O'Donoghue, T., and Rabin, M. 1999. "Doing It Now or Later," *American Economic Review* (89:1), pp. 103-124.
- Pabilonia, S. W., and Zoghi, C. 2005. "Returning to the Returns to Computer Use," *American Economic Review* (95:2), pp. 314-317.
- Papanikos, G. 2015. "The Economic Effects of a Marathon as a Sport Tourism Event," *Athens Journal of Sports* (1:225), pp. 225-240.
- Pereira, P., Duarte, E., Rebelo, F., and Noriega, P. 2014. "A Review of Gamification for Health-Related Contexts," *International Conference of Design, User Experience, and Usability*: Springer, pp. 742-753.

- Pigliapoco, E., and Bogliolo, A. 2008. "The Effects of Psychological Sense of Community in Online and Face-to-Face Academic Courses," *International Journal of Emerging Technologies in Learning (iJET)* (3:4), pp. 60-69.
- Protopsaltis, S., and Baum, S. 2019. "Does Online Education Live up to Its Promise? A Look at the Evidence and Implications for Federal Policy," in: *Center for Educational Policy Evaluation*.
- Raysmith, B. P., Jacobsson, J., Drew, M. K., and Timpka, T. 2019. "What Is Performance? A Scoping Review of Performance Outcomes as Study Endpoints in Athletics," *Sports* (7:3), p. 66.
- Rosenbaum, P. R., and Rubin, D. B. 1983. "The Central Role of the Propensity Score in Observational Studies for Causal Effects," *Biometrika* (70:1), pp. 41-55.
- Rossmann, A., Ranjan, K. R., and Sugathan, P. 2016. "Drivers of User Engagement in Ewom Communication," *Journal of Services Marketing* (30:5), pp. 541-553.
- Saayman, M., and Saayman, A. 2012. "The Economic Impact of the Comrades Marathon," *International Journal of Event and Festival Management* (3:3), pp. 220-235.
- Springer, A. E., Kelder, S. H., and Hoelscher, D. M. 2006. "Social Support, Physical Activity and Sedentary Behavior among 6 Th-Grade Girls: A Cross-Sectional Study," *International Journal of Behavioral Nutrition and Physical Activity* (3:1), p. 8.
- Tello, S. F. 2008. "An Analysis of Student Persistence in Online Education," in *Information Communication Technologies: Concepts, Methodologies, Tools, and Applications*. IGI Global, pp. 1163-1178.
- Terza, J. V., Basu, A., and Rathouz, P. J. 2008. "Two-Stage Residual Inclusion Estimation: Addressing Endogeneity in Health Econometric Modeling," *Journal of Health Economics* (27:3), pp. 531-543.
- Thompson, P. D., Buchner, D., Piña, I. L., Balady, G. J., Williams, M. A., Marcus, B. H., Berra, K., Blair, S. N., Costa, F., and Franklin, B. 2003. "Exercise and Physical Activity in the Prevention and Treatment of Atherosclerotic Cardiovascular Disease: A Statement from the Council on Clinical Cardiology (Subcommittee on Exercise, Rehabilitation, and Prevention) and the Council on Nutrition, Physical Activity, and Metabolism (Subcommittee on Physical Activity)," *Circulation* (107:24), pp. 3109-3116.
- Uetake, K., and Yang, N. 2020. "Inspiration from the "Biggest Loser": Social Interactions in a Weight Loss Program," *Marketing Science* (39:3), pp. 487-499.
- Urtel, M. G. 2008. "Assessing Academic Performance between Traditional and Distance Education Course Formats," *Journal of Educational Technology & Society* (11:1), pp. 322-330.
- Van den Poel, D., and Lariviere, B. 2004. "Customer Attrition Analysis for Financial Services Using Proportional Hazard Models," *European Journal of Operational Research* (157:1), pp. 196-217.
- Viljanen, M., Airola, A., Pahikkala, T., and Heikkonen, J. 2016. "Modelling User Retention in Mobile Games," *2016 IEEE Conference on Computational Intelligence and Games (CIG)*: IEEE, pp. 1-8.
- Weitz, R., Henry, T., and Rosenthal, D. 2014. "Limitations of Nonfinancial Metrics Reported by Social Media Companies," *Journal of International Technology and Information Management* (23:3), p. 2.
- Wiesner, M., Zowalla, R., Suleder, J., Westers, M., and Pobiruchin, M. 2018. "Technology Adoption, Motivational Aspects, and Privacy Concerns of Wearables in the German Running Community: Field Study," *JMIR mHealth and uHealth* (6:12), p. e201.
- Wooldridge, J. M. 2010. *Econometric Analysis of Cross Section and Panel Data*. MIT press.
- Yan, L., and Tan, Y. 2014. "Feeling Blue? Go Online: An Empirical Study of Social Support among Patients," *Information Systems Research* (25:4), pp. 690-709.
- Yeung, S., Yeung, E., and Wong, T. 2001. "Marathon Finishers and Non-Finishers Characteristics. A Preamble to Success," *Journal of Sports Medicine and Physical Fitness* (41:2), pp. 170-176.
- Ying, A., Xu, R., and Murphy, J. 2019. "Two-Stage Residual Inclusion for Survival Data and Competing Risks—an Instrumental Variable Approach with Application to Seer-Medicare Linked Data," *Statistics in Medicine* (38:10), pp. 1775-1801.

Table 1. Variable Definition

Variable	Definition
AliveDays	Number of days between the last login date and user registration date
Female	A user's gender (1 = Female; 0 = Male)
Weight	A user's weight
Height	A user's height
Age	A user's age
VirtualRace	Indicates if a user participates in a virtual race or not
ZodiacMatch (IV)	Indicates if a user's horoscope matches the theme of a virtual race or not
Non-local	Indicates if an individual registered outside Taiwan or not
Attend Running Club	Indicates if a user participates in a running club or not
Mobile User	Indicates if an individual primarily uses mobile devices or not

Table 2-1. Summary Statistics for Sample 1

	Mean	S.D.	Min	Max
AliveDays	213.59	191.49	0.00	675.00
VirtualRace	0.08	0.27	0.00	1.00
Female	0.50	0.50	0.00	1.00
Weight	63.35	10.02	26.00	180.00
Height	169.49	3.98	122.00	200.00
Age	35.23	10.09	17.00	90.00
ZodiacMatch	0.10	0.29	0.00	1.00
Non-local	0.07	0.25	0.00	1.00
Attend Running Club	0.10	0.30	0.00	1.00
Mobile User	0.88	0.32	0.00	1.00

N = 44,391 (Unit: User)

Table 2-2. Summary Statistics for Sample 2

	Mean	S.D.	Min	Max
RunningDistance	16.95	17.20	0.00	76.16
VirtualRace	0.50	0.50	0.00	1.00
Female	0.30	0.46	0.00	1.00
Weight	66.24	11.15	30.00	170.00
Height	169.43	5.87	122.00	200.00
Age	38.90	8.27	17.00	90.00
ZodiacMatch	0.11	0.31	0.00	1.00

N = 222,288 (Unit: User-Week)

Table 3-1. Correlation Table for Sample 1

		1	2	3	4	5	6	7	8	9	10
1	AliveDays	1.00									
2	VirtualRace	0.12*	1.00								
3	Female	-0.14*	-0.06*	1.00							
4	Weight	0.09*	0.01*	-0.46*	1.00						
5	Height	-0.02*	-0.01	-0.27*	0.33*	1.00					
6	Age	0.14*	-0.02*	-0.14*	0.10*	-0.03*	1.00				
7	ZodiacMatch	0.02*	0.18*	-0.01	-0.01	-0.00	-0.01*	1.00			
8	Non-local	-0.07*	-0.02*	-0.01*	-0.03*	0.01*	0.06*	-0.01*	1.00		
9	Attend Running Club	0.10*	0.00	0.01*	0.02*	-0.01*	-0.03*	-0.01*	-0.04*	1.00	
10	Mobile User	-0.16*	-0.08*	0.14*	-0.05*	-0.03*	-0.13*	-0.01	-0.13*	-0.02*	1.00

$N = 44,391$; * $p < .05$

Table 3-2. Correlation Table for Sample 2

		1	2	3	4	5	6	7
1	RunningDistance	1.00						
2	VirtualRace	0.02*	1.00					
3	Female	-0.17*	-0.00	1.00				
4	Weight	-0.00	-0.00	-0.56*	1.00			
5	Height	0.06*	-0.01*	-0.48*	0.52*	1.00		
6	Age	0.22*	-0.00*	-0.11*	0.07*	-0.04*	1.00	
7	ZodiacMatch	-0.02*	0.10*	0.01*	-0.01*	0.00	-0.01*	1.00

$N = 222,288$; * $p < .05$

Table 4. The Impact of a Virtual Race on User Engagement

	(1) AliveDays OLS	(2) Alive3M Probit	(3) AliveDays Tobit
VirtualRace	63.935*** (3.161)	0.467*** (0.025)	39.375*** (2.017)
Female	-13.935*** (1.979)	-0.069*** (0.015)	-3.638*** (1.127)
Weight	1.609*** (0.097)	0.011*** (0.001)	0.934*** (0.061)
Height	-1.957*** (0.227)	-0.013*** (0.002)	-1.151*** (0.130)
Age	2.029*** (0.085)	0.011*** (0.001)	0.764*** (0.049)
_cons	479.821*** (38.411)	1.533*** (0.276)	222.884*** (21.794)
VR fixed effects	Yes	Yes	Yes
N	44391	44391	44391
R^2	0.183		
Log pseudolikelihood		-28184.096	-108673.446

Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table 5. The Impact of a Virtual Race on User Engagement (Matched Sample)

	(1) AliveDays OLS	(2) Alive3M Probit	(3) AliveDays Tobit
VirtualRace	25.036*** (4.669)	0.081** (0.034)	8.051*** (3.077)
Female	-8.289 (5.517)	-0.005 (0.040)	1.996 (3.361)
Weight	1.810*** (0.249)	0.011*** (0.002)	1.075*** (0.178)
Height	-2.965*** (0.510)	-0.012*** (0.004)	-1.151*** (0.377)
Age	3.144*** (0.249)	0.014*** (0.002)	1.153*** (0.160)
_cons	561.416*** (85.648)	1.165* (0.685)	198.129*** (62.552)
VR fixed effects	Yes	Yes	Yes
N	6800	6800	6800
R ²	0.149		
Log pseudolikelihood		-3607.706	-12073.966
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$			

Table 6. Treatment Effect Models to Address Potential Endogeneity

	(1) VirtualRace	AliveDays
VirtualRace		42.769*** (8.779)
ZodiacMatch	0.787*** (0.026)	
Female	-0.216*** (0.024)	-18.542*** (2.284)
Weight	-0.000 (0.001)	1.115*** (0.101)
Height	-0.009*** (0.003)	-1.362*** (0.227)
Age	-0.006*** (0.001)	2.096*** (0.096)
_cons	0.714* (0.432)	431.719*** (38.876)
VR fixed effects	Yes	Yes
N		35104
<i>athrho</i>		0.080*** (0.023)
<i>lnsigma</i>		5.154*** (0.003)
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$		

Table 7. Regression Discontinuity Design with a Hazard Model to Address Potential Endogeneity

	(1)	(2)
	Time to Exit	Time to Exit
VirtualRace	0.480*** (0.102)	0.491*** (0.107)
RunningVar	1.000 (0.001)	1.000 (0.001)
Female		1.121*** (0.045)
Weight		0.987*** (0.002)
Height		1.017*** (0.004)
Age		0.983*** (0.002)
Residual	0.936 (0.191)	0.911 (0.193)
VR fixed effects	No	Yes
<i>N</i>	14140	14140
<i>Log-likelihood</i>	-33023.351	-32874.815

Note: the coefficients are exponentiated coefficients.

Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table 8. Virtual Races Can Be Beneficial for Those with High Social Costs, Those in Greater Need of Social Connections, or Those who Use Mobile Devices (Less Sophisticated Users)

	(1)	(2)	(3)
	AliveDays	AliveDays	AliveDays
VirtualRace	39.743*** (8.759)	44.186*** (8.796)	35.216*** (9.779)
Non-local	-60.187*** (4.000)		
VirtualRace x Non-local	41.150*** (15.225)		
Attend Running Club		67.100*** (2.700)	
VirtualRace x Attend Running Club		14.720* (7.882)	
Mobile User			-73.090*** (2.971)
VirtualRace x Mobile User			29.996*** (7.105)
Female	-19.317*** (2.279)	-18.866*** (2.266)	-21.074*** (2.585)
Weight	1.026*** (0.101)	1.069*** (0.100)	0.056 (0.109)
Height	-1.263*** (0.227)	-1.276*** (0.226)	-0.202 (0.235)
Age	2.176*** (0.095)	2.153*** (0.095)	2.047*** (0.109)
_cons	422.021*** (38.702)	412.539*** (38.708)	452.439*** (40.426)
VR fixed effects	Yes	Yes	Yes
<i>N</i>	35104	35104	22160
<i>athrho</i>	0.080*** (0.023)	0.067*** (0.022)	-0.027 (0.028)
<i>lnsigma</i>	5.150*** (0.003)	5.146*** (0.003)	5.002*** (0.005)

Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table 9. The Impact of a Virtual Race on Daily New Friends

	(1) Daily New Friends (21-day window)	(2) Daily New Friends (9-day window)
VirtualRace	0.000 (0.002)	0.002 (0.002)
Post	-0.012*** (0.002)	-0.003 (0.002)
VirtualRace x Post	0.006* (0.003)	0.007** (0.003)
Female	-0.002 (0.002)	-0.003 (0.002)
Weight	0.000 (0.000)	0.000 (0.000)
Height	0.000 (0.000)	0.000** (0.000)
Age	0.000*** (0.000)	0.000*** (0.000)
_cons	-0.032 (0.030)	-0.092*** (0.035)
VR fixed effects	Yes	Yes
<i>N</i>	285600	122400
<i>R</i> ²	0.002	0.002

Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table 10. How Virtual Races Affect Runners' Performance

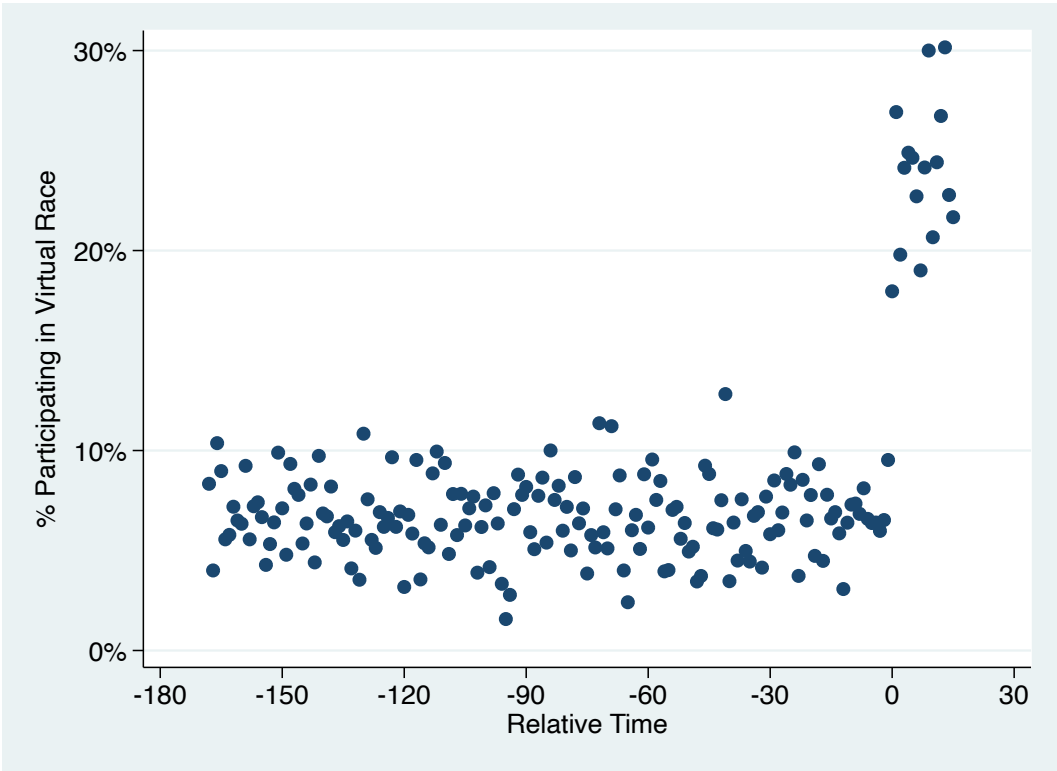
	(1) Running Distance	(2) Running Distance
VirtualRace	-0.005 (0.164)	-0.264 (0.401)
VirtualRace x 3 weeks until VR	-0.239 (0.232)	
VirtualRace x 2 weeks until VR	-0.009 (0.232)	
VirtualRace x 1 week until VR		Omitted
VirtualRace x 0 week since VR	2.817*** (0.242)	
VirtualRace x 1 week since VR	0.710*** (0.238)	
VirtualRace x 2 weeks since VR	0.630*** (0.241)	
VirtualRace x Post		4.405*** (0.601)
Female	-8.093*** (0.090)	-24.279*** (0.389)
Weight	-0.252*** (0.004)	-0.756*** (0.016)
Height	0.141*** (0.007)	0.424*** (0.031)
Age	0.440*** (0.004)	1.321*** (0.018)
_cons	-9.536*** (1.186)	-28.948*** (5.060)
Time (Week or Post) fixed effects	Yes	Yes
VR fixed effects	Yes	Yes
<i>N</i>	222288	74096
<i>R</i> ²	0.095	0.128
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$		

Figure 1. Sample Screenshot of a Virtual Race



比賽時間 2018/7/14 - 2018/7/22	Race Time 2018/7/14 - 2018/7/22
報名結束時間 2018/7/4 23:59:59	Registration End Time 2018/7/4 23:59:59
比賽項目 5K/10K/半程馬拉松/全程馬拉松	Race Item 5K/10K/Half Marathon/Full Marathon
比賽地點 不限地點	Race Location No Constraint

Figure 2. Discontinuity Plot on the Probability of Participating in a Virtual Race



Appendix

Table A1. Univariate Tests before and after Matching

	Before Matching			After Matching		
	Control (N=40,815)	Treated (N=3,576)	Mean Diff	Control (N=3,400)	Treated (N=3,400)	Mean Diff
Female	0.510	0.402	0.108*	0.390	0.413	-0.023
Age	35.280	34.644	0.637*	34.354	34.457	-0.103
Weight	63.320	63.746	-0.427*	63.882	63.641	0.241
Height	169.496	169.383	0.113	169.265	169.394	-0.129
# New Friends Before Virtual Race	0.433	0.896	-0.463*	0.669	0.711	-0.042
Distance Before Virtual Race	12.906	30.412	-17.236*	22.804	23.989	-1.185

* $p < 0.05$

Table A2. IV Tobit Model

	(1) VirtualRace	(2) AliveDays
ZodiacMatch	0.169*** (0.007)	
VirtualRace		24.234** (11.404)
Female	-0.030*** (0.003)	-6.912*** (1.376)
Weight	-0.000 (0.000)	0.690*** (0.062)
Height	-0.001*** (0.000)	-0.901*** (0.142)
Age	-0.001*** (0.000)	0.842*** (0.058)
_cons	0.418*** (0.067)	203.572*** (24.277)
VR fixed effects	Yes	Yes
N	35104	
Log-likelihood	-84142.98	
First-stage F-Statistics	78.40	

Robust standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Table A3. Robustness Checks

	(1)	(2)	(3)	(4)	(5)
	AliveDays	AliveDays	AliveDays	AliveDays	AliveDays
Note	Keep Non-FB Users	Keep Non-Referral Users	Users in Control Group Who Do Not Attend Any Virtual Race	Examine the Impact of “One” Virtual Race on User Engagement	Keep Samples Who Eventually Attend Future Races (After First Possible Virtual Race)
VirtualRace	78.653*** (13.061)	54.552*** (10.366)	71.543*** (7.395)	35.598*** (11.778)	34.809* (17.909)
Female	-0.044 (4.216)	-4.123 (3.550)	-18.150*** (2.335)	-16.937*** (2.354)	-13.337** (5.591)
Weight	0.643*** (0.160)	1.571*** (0.150)	0.964*** (0.103)	1.000*** (0.105)	0.942*** (0.212)
Height	0.146 (0.422)	-1.382*** (0.367)	-1.030*** (0.238)	-0.961*** (0.247)	-1.274*** (0.390)
Age	1.699*** (0.168)	2.048*** (0.146)	1.993*** (0.097)	1.970*** (0.098)	3.262*** (0.260)
_cons	150.350** (72.489)	343.901*** (62.655)	360.051*** (40.626)	342.978*** (42.035)	588.429*** (68.495)
VR fixed effects	Yes	Yes	Yes	Yes	Yes
<i>N</i>	11131	15165	32350	31895	4597
<i>athrho</i>	0.041 (0.029)	0.080*** (0.024)	0.038** (0.018)	-0.192*** (0.032)	-0.136* (0.080)
<i>lnsigma</i>	5.181*** (0.005)	5.203*** (0.004)	5.137*** (0.003)	5.145*** (0.003)	4.927*** (0.016)

Note: The first stage of the endogenous treatment effect model was omitted for brevity
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table A4. Robustness Checks: Alternative Dependent Variables Representing User Engagement and User Spending

	(1) ExerciseDays	(2) ln(# Running Records)	(3) ln(# Log-in Times)	(4) ln(Spending)
VirtualRace	54.051*** (6.701)	0.217*** (0.083)	0.246*** (0.085)	0.395*** (0.055)
Female	-30.740*** (2.270)	-0.406*** (0.023)	-0.252*** (0.020)	-0.179*** (0.021)
Weight	1.003*** (0.104)	0.011*** (0.001)	0.010*** (0.001)	-0.000 (0.001)
Height	-1.990*** (0.242)	-0.030*** (0.002)	-0.017*** (0.002)	-0.001 (0.002)
Age	1.922*** (0.096)	0.025*** (0.001)	0.026*** (0.001)	0.009*** (0.001)
_cons	457.107*** (41.089)	5.766*** (0.419)	3.815*** (0.353)	0.411 (0.399)
VR fixed effects	Yes	Yes	Yes	Yes
<i>N</i>	35104	35104	35104	35104
<i>athrho</i>	0.088*** (0.014)	0.267*** (0.023)	0.250*** (0.028)	0.027*** (0.008)
<i>lnsigma</i>	5.184*** (0.003)	0.594*** (0.003)	0.440*** (0.003)	0.536*** (0.012)

Note: The first stage of the endogenous treatment effect model was omitted for brevity
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table A5. The Impact of a Virtual Race on User Engagement for Users Who Do Not Have Social Connections Prior to Virtual Races

	(1) AliveDays
VirtualRace	44.546*** (10.983)
Female	-4.426 (3.739)
Weight	1.664*** (0.163)
Height	-1.511*** (0.420)
Age	1.905*** (0.154)
_cons	342.600*** (71.588)
VR fixed effects	Yes
<i>N</i>	11131
<i>athrho</i>	0.056** (0.024)
<i>lnsigma</i>	5.189*** (0.005)

Note: The first stage of the endogenous treatment effect model was omitted for brevity
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table A6. Parallel Assumption on Daily New Friends

	(1) Daily New Friends
VirtualRace	0.012 (0.029)
Pre-Treatment (Week 1)	0.115*** (0.041)
Pre-Treatment (Week 3)	-0.052** (0.026)
VirtualRace x Pre-Treatment (Week 1)	-0.036 (0.051)
VirtualRace x Pre-Treatment (Week 3)	0.004 (0.036)
Female	-0.006 (0.020)
Weight	0.001 (0.001)
Height	0.001 (0.002)
Age	0.001 (0.001)
_cons	-0.223 (0.394)
VR fixed effects	Yes
<i>N</i>	20250
<i>R</i> ²	0.016

Note: Pre-Treatment (Week 2) is the omitted baseline group.

Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table A7. Intention to Run Next the Virtual Race

	(1) Next VirtualRace	(2) Next VirtualRace
	Sample 1	Sample 2
VirtualRace	1.940*** (0.027)	2.520*** (0.021)
Female	-0.081*** (0.029)	-0.035 (0.025)
Weight	0.011*** (0.001)	0.002** (0.001)
Height	-0.021*** (0.003)	-0.003 (0.002)
Age	0.005*** (0.001)	0.002** (0.001)
_cons	0.455 (0.546)	-1.876*** (0.313)
VR fixed effects	Yes	Yes
<i>N</i>	41684	34864
<i>Log Likelihood</i>	-6025.098	-12293.817

Robust standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Table A8. How Virtual Race Affects the Likelihood to Attend Offline Races

	(1) Attend Any Offline Race	(2) Attend Any Offline Race
VirtualRace	0.142*** (0.009)	0.596*** (0.010)
Female	-0.030*** (0.004)	-0.009 (0.008)
Weight	0.001*** (0.000)	-0.001*** (0.000)
Height	-0.003*** (0.000)	0.002** (0.001)
Age	0.002*** (0.000)	0.002*** (0.000)
_cons	0.514*** (0.078)	-0.172 (0.130)
VR fixed effects	Yes	Yes
<i>N</i>	35104	16001
<i>athrho</i>	0.051*** (0.007)	-1.164*** (0.022)
<i>lnsigma</i>	-1.240*** (0.007)	-0.914*** (0.008)

Note: The first stage of the endogenous treatment effect model was omitted for brevity
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Table A9. Univariate Tests Before and After Matching

	Before Matching			After Matching		
	Control (N=492,859)	Treated (N=73,675)	Mean Diff	Control (N=18,524)	Treated (N=18,524)	Mean Diff
Female	0.274	0.283	-0.009*	0.302	0.299	0.003
Age	39.49	39.305	0.184*	38.941	38.863	0.078
Weight	66.154	66.473	-0.319*	66.246	66.226	0.020
Height	169.667	169.357	0.310*	169.463	169.39	0.072
Running Distance (Pre-treatment 3 Weeks)	54.253	79.72	-25.467*	51.182	50.705	0.477

* $p < .05$

Table A10. Treatment Effect Models to Address Potential Endogeneity

	(1) VirtualRace	(2) Running Distance
ZodiacMatch	0.408*** (0.009)	
VirtualRace		-4.871*** (0.673)
VirtualRace x 3 weeks until VR		-0.325 (0.253)
VirtualRace x 2 weeks until VR		0.036 (0.253)
VirtualRace x 1 week until VR		Omitted
VirtualRace x 0 week since VR		2.816*** (0.264)
VirtualRace x 1 week since VR		0.651** (0.259)
VirtualRace x 2 weeks since VR		0.545** (0.263)
Female	0.013 (0.008)	-8.291*** (0.103)
Weight	-0.000 (0.000)	-0.278*** (0.004)
Height	-0.001 (0.001)	0.160*** (0.008)
Age	-0.001** (0.000)	0.427*** (0.005)
Week fixed effects	Yes	Yes
VR fixed effects	Yes	Yes
<i>N</i>		188250
<i>athrho</i>		0.183*** (0.025)
<i>lnsigma</i>		2.809*** (0.004)

Note: The first stage of the endogenous treatment effect model was omitted for brevity
Robust standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$

Figure A1. Key Features of the *RunningPlatform*

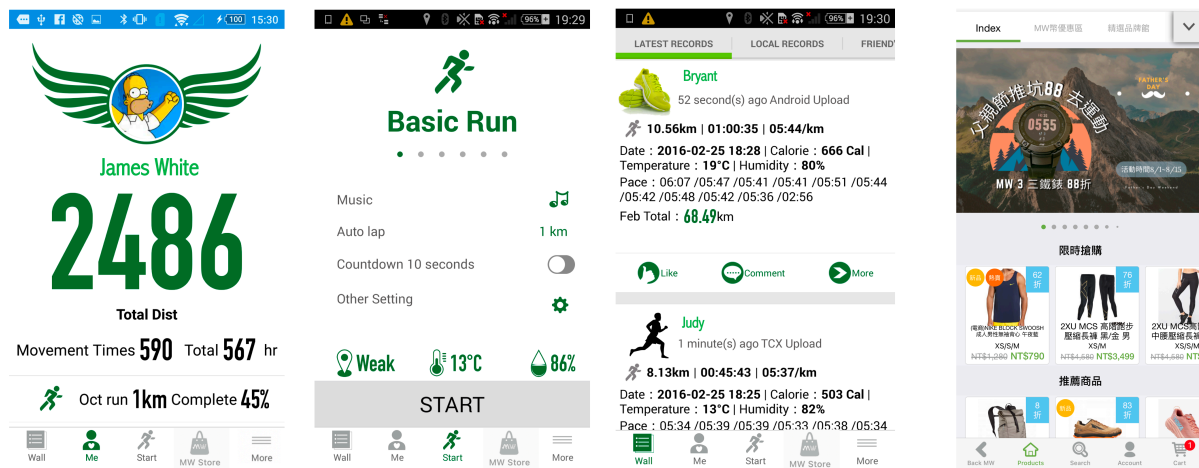


Figure A2. Average 14-day Running Distance before and after the Introduction of Virtual Race

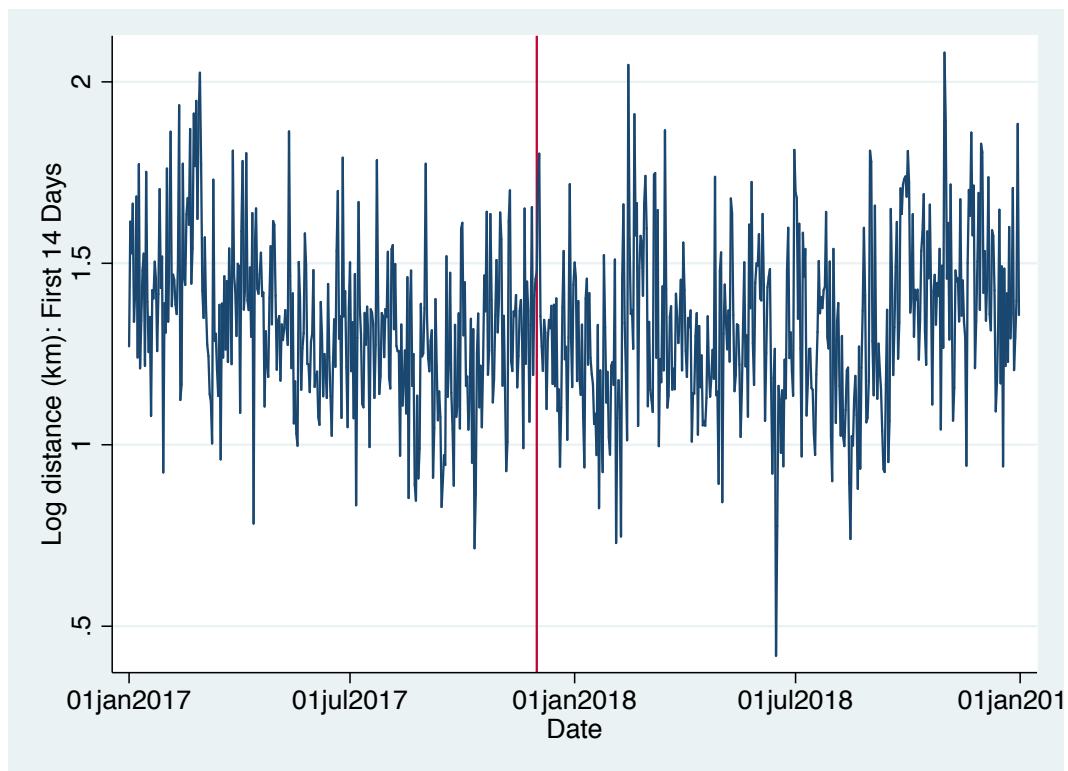


Figure A3. Distributions of Days from Registration to the First Virtual Race Start Date between Participants and Non-Participants

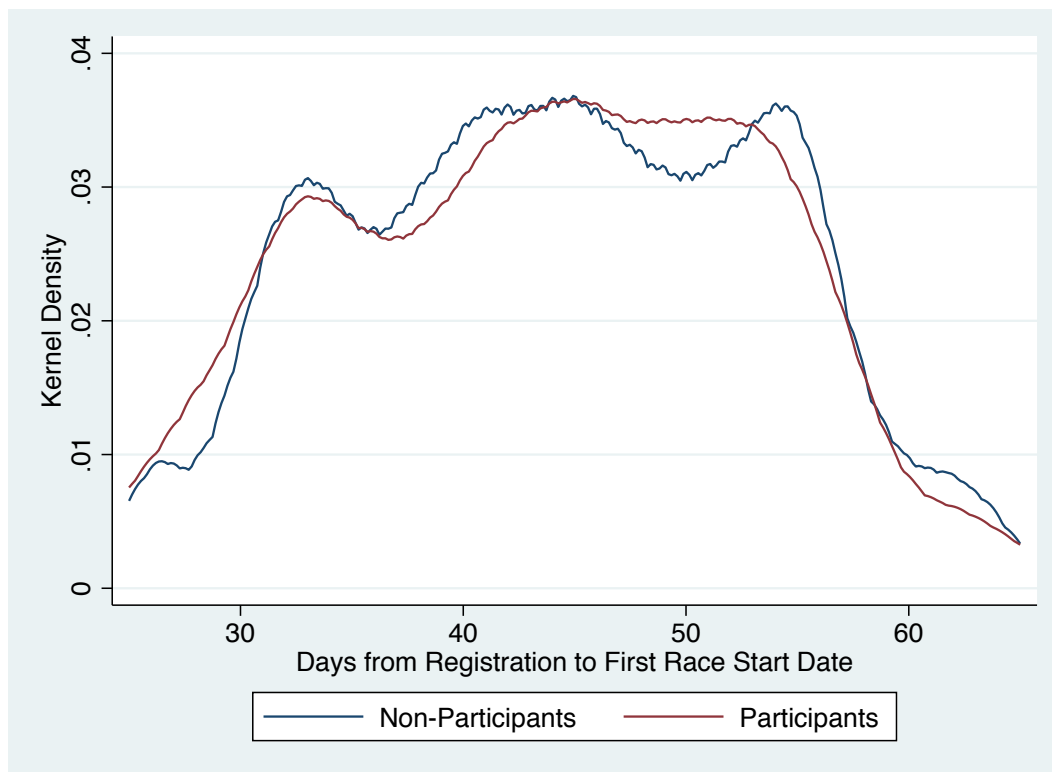


Table B1. Apps or Platforms that Provide Regular Participation of Virtual Races

Apps or platforms	Virtual Race	Short Description and Frequencies
Run Disney	RunDisney Virtual Race Event	RunDisney Virtual Race is “a running event that you do on your own time in the location of your choice. Once you register, you’ll need to finish the distance at any interval or pace within the completion timeframe designated for your Virtual Race. Afterwards, you’ll receive your RunDisney Virtual Race medal in the mail. ³⁷ ” Starting from 2016, the virtual races have attracted hundreds of thousands of participants to attend those events each year. In addition, the RunDisney Virtual Race was typically hosted with a theme. For instance, between June and August 2022, the virtual race’s theme was “FROZEN.” For the February event in 2023, the theme of the virtual race was “Disney Princess.”
Virtual Run Events	Virtual Run Events: Virtual Race	A virtual race in the Virtual Run Events “is a race that can be run (or walked) from any location you choose. You can run, jog, or walk on the road, on the trail, on the treadmill, at the gym, or on the track (or even at another race). You get to run your own race, at your own pace, and time it yourself. And your medal will be shipped directly to you. Complete a virtual 5K, 10K, half marathon and more! ³⁸ ” In addition, Virtual race in the Virtual Run Events platform has helped to raise more than \$926,413 for charities. The virtual race is hosted frequently. For instance, there were 14 virtual races held in September 2022 alone.
Joyrun	Joyrun Virtual Race	Joyrun is a leading running app in China that hosts marathon races for both online and offline. Users need to sign up for a virtual race before it starts. After the deadline of registration, runners need to complete the virtual race in a designated time window. The mobile app or other appointed apps is required to trace the running details. As a leader of the virtual race, Joyrun started the first virtual marathon in January 2015. It has attracted more than 110 million registered users that cover more than 2,100 cities worldwide ³⁹ .
Codoon	Codoon Online Race	Codoon is a leading outdoor sports platform that has held more than 1,000 online and offline races since 2016. Users have to use the mobile application provided by Codoon to complete a race in any location within a designated time window. According to a recent report ⁴⁰ , Codoon hosted 199 virtual events in the year 2021, and attracted 210,000 runners per race, with an average of 78.13% of completion rate.
VirtualRace.org	atlasGO Virtual Race	VirtualRace.org hosts virtual races, that “is a race that you run on an app! You don’t go to a physical event. You can run wherever you want and whenever you want. You are the boss of your own race. You can run in your neighborhood, in the park, on a treadmill,... You can run alone, with your friends, with your dog,... your choice! Once you bought your ticket you will receive a code to join the race on the app and GO! ⁴¹ ” Runners must use the tracking app (atlasGO app) to report their running details. VirtualRace.org hosts virtual races on a continual basis. In 2021 alone, it hosted 43 virtual races on the platform.
Social Media Platform	Facebook groups for “Virtual Run Events” or Eventbrite events for “Virtual Marathon”	Digital platforms such as Facebook or Everbrite provide runners with a place to host virtual races. For instance, there are over 250 active groups for virtual races on Facebook, and over 10,000 virtual races can be found on Eventbrite. For these platforms, most virtual races restrict a time window for the runners to complete the running. While tracking technology is not mandatory, runners need to self-report the results of their performance to the platform.

³⁷ <https://www.rundisney.com/faq/>

³⁸ <https://www.virtualrunevents.com/how-it-works/>

³⁹ <https://www.thejoyrun.com/about.php>

⁴⁰ https://www.sohu.com/a/513529550_120070669

⁴¹ <https://virtualrace.org/>

Table B2. Summary of Related Studies in Wearable Devices, mHealth, or Online Health Platform

Author	Treatment/ Key Independent Variable(s)	Key outcome variable(s)	Main Findings
Adjerid et al. (2021)	Economic incentives + Gain-Loss incentives	Step counts	There was a modest increase in physical activities for individuals who were provided with the choice of gain-loss incentives in an experimental setting, and a significant and persistent increase for those assigned (i.e., against users' preferences) to the same gain-loss incentives. Importantly, the persistent gains were driven by individuals who utilized the wearable's goal-setting tool.
Aral and Nicolaidis (2017)	Social contagion	Physical activity	Leveraging weather patterns across geographies, the study quantifies the contagion effects in exercise behaviors across a global social network. The authors find that the embeddedness and structural diversity theories of social contagion can explain the influences observed in this study, but the complex contagion theory does not.
Bavafa et al. (2018)	E-visits (secure messaging)	Office visits and patients' health indicators	E-visits (secure messaging) trigger 6% more office visits, but also cause 15% fewer new patients' acceptance.
Hagen et al. (2021)	Freemium and premium	User engagement and retention	Adoption of premium in a mHealth system leads to an increase in user engagement and retention.
Huang et al. (2019)	Structural, relational, and cognitive dimensions of social capital	Provision of informational and emotional support	All three dimensions of social capital (i.e., structural, relational, and cognitive social capital) positively increase the provision of emotional support, but structural and relational social capital facilitate engagement in companionship; and only cognitive social capital increases the provision of informational support.
Ghose et al. (2021)	mHealth apps	Health and economic impact	mHealth app adoption increases both short-term metrics (blood glucose and glycated hemoglobin levels) and longer-term metrics (hospital visits and medical expenses). Interestingly, personalized mobile messages have an unexpected effect on patients' app engagement, leading to lower health improvement.
Kato-Lin et al. (2016)	Mobile-based visual diary, image-based dietitian support, and peer engagement	User engagement	A mobile-based visual diary and dietitian support increase user engagement by 43.8% and 50.7%, but peer engagement leads to a negative effect by decreasing engagement by 33.8%.
Liu et al. (2021)	Reciprocity	Physical activities	Reciprocity-based incentives on a mobile-enabled physical activity platform can outperform self-interest incentives for motivating physical activities, and this effect is contingent on the social closeness between senders and receivers.
Müller et al. (2016)	SMS	Exercise frequency	SMS text messaging can be an effective means to promote exercise in older adults from an upper-middle-income country, but the effects were not maintained when SMS text messaging ceased.

Yan and Tan (2014)	Social support	Mental health	Patients can benefit from learning from others in an online community. In addition, the authors find that information support is the most prevalent type of support in the online healthcare community, but emotional support is the most effective form to help patients move to a healthier state.
Zhou et al. (2021)	Self-regulation focuses	Engagement	Users in an online weight-loss community increase their recording behaviors and decrease their self-weighing behaviors after they participate in online social activities.
Kwon et al. (2022)	Mobile app + offline office visit	Weight loss	The mobile app channel and the offline channel substitute each other in the short-term weight loss program. In the long run, only the mobile app is effective for the sustainability of weight loss. The results can be explained by self-regulation enabled by mobile app usage.
Fan et al. (2022)	Online consultation services	Offline appointments of doctors	The number of offline doctor appointments increased after the opening of the online consultation service. The effect is heterogeneous based on doctor-related factors, such as online rating, hospital ranking, and doctor title.
This study	Virtual race	New user engagement	A virtual race can increase the tenure of new users by 42.8 days. This study also finds that social interactions enabled by the digital platform play a critical role in realizing the benefits of event virtualization. Further analysis reveals that virtual races can encourage new users to participate in more offline races. Finally, this study shows that existing users who participate in a virtual race outrun those in the control group by 4.41 km in a three-week period.

References

- Adjerid, I., Loewenstein, G., Purta, R., and Striegel, A. 2021. "Gain-Loss Incentives and Physical Activity: The Role of Choice and Wearable Health Tools," *Management Science* (64:4), pp. 2642–2667.
- Aral, S., and Nicolaides, C. 2017. "Exercise Contagion in a Global Social Network," *Nature Communications* (8), p. 14753.
- Bavafa, H., Hitt, L. M., and Terwiesch, C. 2018. "The Impact of E-Visits on Visit Frequencies and Patient Health: Evidence from Primary Care," *Management Science* (64:12), pp. 5461-5480.
- Fan, W., Zhou, Q., Qiu, L., and Kumar, S. 2022. "Should Doctors Open Online Consultation Services? An Empirical Investigation of Their Impact on Offline Appointments," *Information Systems Research* (Articles in Advance).
- Ghose, A., Guo, X., Li, B., and Dang, Y. 2021. "Empowering Patients Using Smart Mobile Health Platforms: Evidence from a Randomized Field Experiment," *MIS Quarterly* (46:1), pp. 151-192.
- Hagen, L., Jiang, Y., Knäuper, B., Uetake, K., and Yang, N. 2021. "The Implications of Premium Version Adoption on Customer Relationship Management: An Empirical Study of Mhealth User Engagement and Retention," in: *Available at SSRN 3573193*.
- Huang, K.-Y., Chengalur-Smith, I., and Pinsonneault, A. 2019. "Sharing Is Caring: Social Support Provision and Companionship Activities in Healthcare Virtual Support Communities," *MIS Quarterly* (43:2), pp. 395-424.
- Kato-Lin, Y.-C., Abhishek, V., Downs, J. S., and Padman, R. 2016. "Food for Thought: The Impact of M-Health Enabled Interventions on Eating Behavior," in: *SSRN 2736792*.

- Kwon, H. E., Dewan, S., Oh, W., and Kim, T. 2022. "Self-Regulation and External Influence: The Relative Efficacy of Mobile Apps and Offline Channels for Personal Weight Management," *Information Systems Research* (Articles in Advance).
- Liu, C.-W., Gao, G. G., and Agarwal, R. 2021. "Reciprocity or Self-Interest? Leveraging Digital Social Connections for Healthy Behavior," *MIS Quarterly* (46:1), pp. 261-298.
- Müller, A. M., Khoo, S., and Morris, T. 2016. "Text Messaging for Exercise Promotion in Older Adults from an Upper-Middle-Income Country: Randomized Controlled Trial," *Journal of Medical Internet Research* (18:1), p. e5.
- Yan, L., and Tan, Y. 2014. "Feeling Blue? Go Online: An Empirical Study of Social Support among Patients," *Information Systems Research* (25:4), pp. 690-709.
- Zhou, T., Yan, L., Wang, Y., and Tan, Y. 2021. "Turn Your Online Weight Management from Zero to Hero: A Multidimensional, Continuous-Time Evaluation," *Management Science* (68:5), pp. 3507-3527.